When Not to Trade:

Drawdown Regimes and the Implementation of Large-Cap Momentum

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Abstract

A large-cap, long-only momentum portfolio earns essentially all of its benchmark-beating return in a minority of months: those a real-time regime signal, filtered from the market drawdown alone, classifies as panic. We show this with a two-stage framework in which a Bayesian hidden Markov model turns the drawdown into a continuous panic probability and a gradient-boosted model conditions stock selection on that probability jointly with twelve momentum horizons, evaluated on an expanding out-of-sample walk over the point-in-time 1,000 largest U.S. stocks (2011–2025, extended back to 2000). The honest headline is negative: the strategy earns no unconditional risk-adjusted alpha in this configuration. Its gross active return concentrates in the quarter of months classified as panic (+0.72% versus +0.04% per month in calm), where selection inverts from past winners to beaten-down recovery candidates ranked, on average, below the cross-sectional average at every momentum horizon. The signal's practical product is a when-to-trade rule. Roughly three-quarters of a monthly-rebalanced book's turnover occurs in calm months that earn nothing, so conditioning a buy/hold trading band on the regime, wide in calm and tight in panic, raises the net information ratio at 10 bp costs from 0.06 under monthly rebalancing to 0.39, against 0.29 for the best static band, and the breakeven cost from 14 to 42 basis points, a statistically significant improvement over monthly rebalancing of the same signal. To our knowledge this is the first state-dependent buy/hold trading band implemented on a cross-sectional selection portfolio.

Contents

1 Introduction

Cross-sectional momentum[2] presents a practitioner with an awkward bundle. In its investable form, a long-only tilt within liquid large capitalizations, the raw signal is cheap to hold but expensive to trade, its performance is dominated by a small number of episodes, and its famous failure mode arrives exactly when markets rebound from a crash (Daniel and Moskowitz, 2016). Transaction-cost studies have made the stakes concrete: many published anomalies do not survive realistic trading costs (Novy-Marx and Velikov, 2016; Detzel et al., 2023), and the machine-learning strategies that dominate recent academic horse races are among the most cost-fragile (Azevedo et al., 2024). The question this paper asks is therefore not whether momentum "works," but a sharper, implementational one: *when* is a large-cap, long-only momentum book worth tilting away from its benchmark, and when is trading it worth paying for?

The crash-management literature answers a different question. Its instruments operate on the strategy's exposure or its signal weights: scaling by realized volatility (Barroso and Santa-Clara, 2015; Moreira and Muir, 2017), gating on a binary market state (Cooper et al., 2004), or blending two momentum horizons by market cycle for a single index (Goulding et al., 2023); none conditions a cross-sectional selection rule on the regime, and the regime variable never meets the trading-cost decision. On the cost side, mitigation techniques such as buy/hold trading bands are studied as static policies (Novy-Marx and Velikov, 2016, 2019), and the theory that anticipates state-dependent no-trade regions (Jang et al., 2007; Lynch and Tan, 2011) has, to our knowledge, not been taken to a cross-sectional selection portfolio; the closest empirical antecedent conditions trading speed, not a no-trade region, in a market-timing setting (Collin-Dufresne et al., 2020). The gap between these two literatures, a regime signal that decides both which stocks to hold and when trading is worth its cost, is where this paper operates.

Research question

When does a large-cap, long-only momentum portfolio earn its active return over the benchmark, and how should the answer change what, and when, the portfolio trades?

Approach

A two-stage framework answers the question. First, a Bayesian hidden Markov model estimated on a single macrofinancial series, the market drawdown, produces a continuous filtered probability of market panic that updates in real time; the momentum-crash literature identifies the market's decline-and-recovery state as the conditioning information that matters, and a train-only selection rule never adds a second stress feature (Section 4.1). Second, an XGBoost model combines that probability with twelve momentum horizons to

[2] Cross-sectional momentum ranks stocks against each other at each point in time and holds the relative winners (Jegadeesh and Titman, 1993).

rank the point-in-time 1,000 largest U.S. stocks into a long-only, value-weighted top-decile portfolio, held against the value-weighted top-1000 benchmark. Everything is evaluated on an expanding yearly walk, out-of-sample over 2011–2025 and extended back to 2000, with transaction costs treated as a first-class object rather than a robustness footnote.

Headline findings

The honest headline is negative: in this investable configuration the strategy earns no unconditional risk-adjusted alpha (six-factor alpha +1.6%, $t = 0.34$) and its Sharpe ratio trails the benchmark's. What survives is structure. The strategy's active return is concentrated almost entirely in the roughly one quarter of months the regime signal classifies as panic (+0.72% per month, against +0.04% in calm), and in those months the model's selection inverts: the long portfolio is rebuilt from beaten-down recovery candidates that rank, on average, below the cross-sectional average at every momentum horizon, a basket no single-horizon winner-ranked rule would assemble. Comparators show the regime signal's distinct contribution is calm-month discipline, keeping the book benchmark-like when the tilt earns nothing, rather than a larger panic payoff. That observation converts into the paper's deliverable: a regime-conditional trading band, wide in calm and tight in panic, that raises the net information ratio at 10 bp costs from 0.06 to 0.39, where the best static band reaches 0.29, and the breakeven cost from 14 to 42 basis points, a statistically significant improvement over monthly rebalancing of the same signal (the lead over the static band is consistent across cost levels but not individually significant). The extended backtest adds the warning label, a dot-com episode in which the reversal bet loses more than the index, and international replication shows the mechanism travels in parts: the value-where-momentum-fails result appears in Japan, the panic-concentration pattern in the UK.

Contributions

The paper makes three contributions. First, the methodological combination of a continuous probabilistic regime filter with a nonlinear cross-sectional model, in its most parsimonious form: a univariate HMM on market drawdown routing an XGBoost selection rule. Second, an empirical characterization of how the momentum cross-section reorganizes across regimes at the level of individual monthly picks, via a four-cluster decomposition of the model's own selections. Third, the regime-conditional trading band: to our knowledge the first state-dependent buy/hold band implemented on a cross-sectional selection portfolio, connecting the regime-conditional momentum literature to the transaction-cost literature it has never let govern its trading.

Outline

Section 2 reviews the momentum, regime, machine-learning, and transaction-cost literatures this paper draws on. Section 3 develops the two-stage framework. Section 4 describes the sample and makes the case for drawdown as the sole regime input. Section 5 presents the out-of-sample results: performance and its regime structure, the selection mechanism, robustness including the 2000–2025 extension and international replication, and the transaction-cost analysis that delivers the when-to-trade rule. Section 6 concludes with limitations and extensions.

2 Literature Review

2.1 The Momentum Anomaly and Its Horizon Structure

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets. Portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. They examine sixteen formation-and-holding-period combinations from three to twelve months, with the twelve-month formation period emerging as the most profitable. The strategy has since been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanism behind momentum remains debated. Behavioral explanations include gradual information diffusion (Hong and Stein, 1999) in addition to the joint conservatism-representativeness mechanism of Barberis et al. (1998), while risk-based explanations link momentum to time-varying systematic risk. This paper does not take a position on the underlying mechanism but examines the empirical regularity that momentum's cross-sectional structure varies with market conditions.

A defining feature of momentum is its sensitivity to the lookback horizon. Short lookbacks of one to three months capture short-term reversal effects (Jegadeesh, 1990; Lehmann, 1990), intermediate horizons (months 7 to 12 prior to the formation month) capture continuation (Novy-Marx, 2012), and lookbacks beyond twelve months reflect long-term mean reversion (De Bondt and Thaler, 1985). Lee and Swaminathan (2000) document a momentum lifecycle in which the same signal evolves from continuation to reversal as information ages. Together these findings suggest that momentum is a family of horizon-specific patterns whose relative strength varies with market conditions, not a single signal. The framework here tests this directly by exposing all twelve monthly horizons jointly to the cross-sectional model rather than committing to a single lookback ex ante.

2.2 Momentum Crashes and Exposure-Based Solutions

Despite its long-run profitability, momentum is prone to severe crashes. Daniel and Moskowitz (2016) document that momentum portfolios suffered catastrophic losses during the 1932 and 2009 market rebounds, and show that these crashes are predictable. When the market has recently declined, past losers tend to be high-beta stocks that rally disproportionately during recoveries, while past winners are low-beta defensive stocks that lag. In economic terms, momentum is a bet that the recent regime persists. It earns the continuation premium when the regime holds, and crashes at regime transitions, when the cross-sectional ordering inherited from the prior regime becomes the wrong bet for the new one. Barroso and Santa-Clara (2015) propose a simple risk-management response, scaling momentum exposure inversely with the strategy's recent realized volatility, a constant-volatility overlay that substantially raises the Sharpe ratio. Bianchi et al. (2024) pursue a related approach by constructing a crash indicator from the joint time variation of conditional volatility and skewness, again operating on the strategy's exposure rather than its composition. This paper explores a complementary composition-side route: rather than scaling exposure during stress, the model re-ranks the cross-section conditional on the regime, so the fully-invested portfolio changes what it holds rather than how much it holds.

The liquidity literature provides a theoretical basis for why momentum crashes reverse. Margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, the mechanism central to the regime-momentum interpretation developed below.

2.3 Regime Models and Regime-Conditional Asset Pricing

Hidden Markov Models have a long history in econometrics. The probabilistic machinery dates to Baum et al. (1970); Hamilton (1989) introduced the regime-switching tradition with a business-cycle application to U.S. GNP growth, and Bayesian estimation via Gibbs sampling was developed by Albert and Chib (1993) with the multivariate-emission generalization now standard (Frühwirth-Schnatter, 2006). In the spirit of Hamilton's approach of fitting the regime model to macroeconomic observables rather than to returns directly, this paper fits the regime model to a macrofinancial stress observable, the market drawdown.

A separate literature asks whether asset-pricing anomalies are themselves regime-conditional. Cooper et al. (2004) establish that momentum profits depend on the prior market state, with strong continuation following up-markets and weaker or reversed performance following down-markets. More recent work integrates regime estimation into portfolio construction, for example Shu and Mulvey (2024), who feed sparse-jump-model factor regimes into a Black-Litterman allocation.

The most directly related work is Goulding et al. (2023), who classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) from the trailing one-month and rolling twelve-month market returns and blend the two momentum horizons via a cycle-specific weight $\alpha \cdot mom_1 + (1-\alpha) \cdot mom_{12}$. The framework here generalizes that approach along three axes: from a deterministic four-state classification based on trailing returns to a continuous panic probability from a Bayesian HMM on a single real-time stress observable, the market drawdown; from two horizons to the full twelve-horizon term structure; and from a parametric linear blend to a learned nonlinear combination.

Their strategies are a time-series optimization of a single market index rather than a cross-sectional stock-selection portfolio, so a direct performance comparison is not appropriate.

The unifying feature of this subsection is that the regime variable enters portfolio decisions through a hand-coded rule, threshold, or parametric weight, never as a continuous learned feature. This paper instead supplies the probability of being in a calm/panic regime as a feature to a nonlinear cross-sectional model, which learns how to use it to optimize portfolio construction.

2.4 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly. Gu et al. (2020) compare a wide range of methods on a kitchen-sink characteristic set including multiple momentum horizons, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinear interactions among firm characteristics. Gradient-boosted trees (Chen and Guestrin, 2016) have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and interactions without explicit specification, and are robust to irrelevant features. A central challenge in applying these methods to asset pricing is interpretability. This paper addresses it on two fronts: SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020) attribute each prediction to feature-level contributions, and a clustering of the portfolio's monthly cross-sectional z-curves into recurring selection modes characterizes what the model selects in each regime rather than only which features matter.

The closest methodological precedent is Beckmeyer and Wiedemann (2025), who use machine learning to learn data-driven weights across past daily returns for a single momentum signal, constructing a dynamically reweighted strategy that outperforms the conventional twelve-minus-one specification. The framework here differs in two respects. First, the weighting is learned not over the daily history of one signal but over a vector of twelve monthly horizons. Second, the weighting is learned *conditional on a regime context variable*, in contrast to the unconditional weighting in Beckmeyer and Wiedemann (2025), which averages across calm and panic regimes. Isolating the regime-conditioning role is this framework's primary methodological point, and the paper tests whether disaggregating that average reveals a structural reorganization of the term structure between states.

2.5 Transaction Costs and Implementation

A strategy claim is only as good as its net-of-cost version. Novy-Marx and Velikov (2016) tax the anomaly literature with realistic effective spreads and find that many published strategies lose most or all of their paper returns, with momentum among the costliest to trade at roughly 65 bp per month; Detzel et al. (2023) make the same point at the level of asset-pricing model comparison, arguing that models should be judged net of implementation costs. The standard mitigation toolkit is static: Novy-Marx and Velikov (2016) find the buy/hold trading band, which enters positions at an extreme signal rank but holds them until the rank decays through a wider exit threshold, the single most effective simple cost-mitigation technique, and Novy-Marx and Velikov (2019) confirm in a head-to-head comparison that banding beats reduced rebalancing frequency and low-cost-universe restriction because it preserves exposure to the selection signal. The stakes are highest exactly where this paper operates: Azevedo et al. (2024) report that machine-learning strategies, despite the largest gross returns in recent comparisons, are generally not rescued by static cost-mitigation.

A separate theoretical literature derives *state-dependent* trading policies. Jang et al. (2007) show that with regime-switching investment opportunities the optimal no-trade region itself depends on the regime; Lynch and Tan (2011) calibrate portfolio choice with state-dependent transaction costs; Gârleanu and Pedersen (2013) derive the aim-portfolio partial-adjustment rule under quadratic costs, with an unconditional trading speed. The closest empirical antecedent is Collin-Dufresne et al. (2020), who condition trading *speed* on a liquidity regime in a market-timing setting. What none of this literature provides is an empirical implementation of a state-dependent buy/hold band on a cross-sectional stock-selection portfolio: the execution-side contribution of this paper sits precisely at that junction, conditioning the band that the static-mitigation literature recommends on the regime that the momentum literature says matters.

Despite these advances, to the best of our knowledge no published study examines how the full term structure of cross-sectional momentum predictability reorganizes across regimes, whether capturing this reorganization requires nonlinear modeling, or how a regime signal should govern the trading of the resulting portfolio.

3 Methodology

This section combines ideas and statistical methods from the literatures reviewed in Section 2 into a two-stage framework that tests the research question posed in Section 1. Stage 1 estimates a Bayesian two-state Hidden Markov Model on a single macrofinancial stress indicator, the market drawdown, producing a continuous probability of the panic state. Stage 2 combines that probability with cross-sectional momentum inside a series of stock-selection models. The case for conditioning on drawdown alone, grounded in the momentum-crash literature and confirmed by a selection rule that uses training information only, is developed in Section 4.1.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This study employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes[3]. The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t, where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. The labels *calm* and *panic* reflect the economic nature of the chosen stress feature, which is diagnostic of market stress rather than trend direction, business-cycle phase, or volatility alone. An HMM fitted to a different feature set would identify different regimes. As emphasized by Hamilton (1989), market regimes are not directly observable. An investor cannot know with certainty whether the market is currently in a calm or panic state, and classifications that look obvious in retrospect are genuinely difficult to identify in real time. The HMM formalizes this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data.

Observation equation

Conditional on the regime, the vector of standardized features[4] $\mathbf{z}_t \in \mathbb{R}^D$ is drawn from a regime-specific multivariate normal distribution. We assume that, conditional on the regime $s_t = k$, $\mathbf{z}_t$ is normally distributed with mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$ for $k \in \{0, 1\}$ (calm and panic). Writing $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the multivariate normal distribution with the indicated mean and covariance, and using the shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t)$

[3] The choice of $K = 2$ is validated empirically. Appendix G.5 shows that increasing to $K = 3, 4,$ or 5 states does not improve downstream portfolio performance and introduces estimation instability.

[4] $\mathbb{R}^D$ denotes the D-dimensional space of real-valued vectors; the D stress indicators at month t are stacked into a single point in this space so the HMM can model them jointly. The framework is stated for general D because the selection exercise of Section 4.1 also estimates multi-feature candidate specifications. The production model uses $D = 1$ (market drawdown only), so the emission distributions reduce to univariate normals.

for the history of observations through month t,

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \qquad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1). \tag{1}$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D\times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. For the multi-feature candidate specifications considered during selection, the multivariate form captures both level differences in individual features across regimes and changes in the correlation structure. In the drawdown-only production model ($D = 1$) it reduces to a regime-specific mean and variance: panic is identified as the state with a deeper average drawdown and more volatile drawdown innovations. The Gaussian assumption is standard in the HMM literature. Appendix H.8 verifies its adequacy by comparing it to heavier-tailed multivariate Student-t emissions.

Transition equation

The evolution of the hidden state follows a Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \tag{2}$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month. The expected duration of regime k is $1/(1 - P_{kk})$ months, so a persistence of 0.95 corresponds to a regime that lasts roughly 20 months on average.

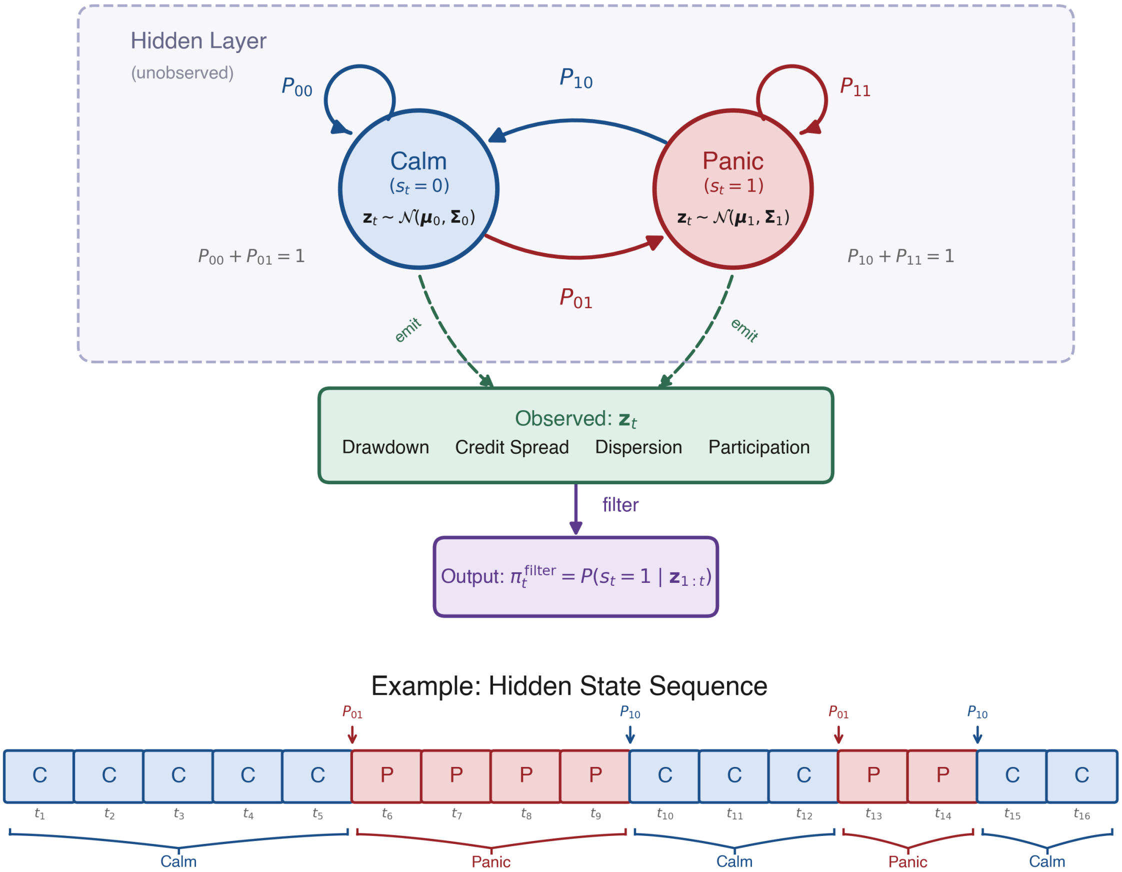

Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the observation vector $\mathbf{z}_t \in \mathbb{R}^D$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximization (Dempster et al., 1977), which finds maximum likelihood point estimates.

Maximum likelihood estimation of regime-switching HMMs has several known shortcomings for the present application. The likelihood surface is multimodal, with at minimum two equivalent global optima arising from the invariance of the likelihood under state relabeling. Additionally, EM can converge to degenerate solutions in which one regime collapses around a single observation (Frühwirth-Schnatter, 2006). Both problems are exacerbated for rare regimes. The asymptotic standard errors that frequentist inference relies on become unreliable when few observations support the rare-state estimates, and the standard regularity conditions can fail outright: the persistence parameters P_{00} and P_{11} are unidentified when the two regimes' means and covariances coincide, and the information matrix becomes singular at that point (Hamilton, 1989, footnote 12).

The Bayesian approach addresses these issues directly. Posterior sampling integrates over parameter uncertainty rather than relying on a point estimate. Conjugate priors regularize rare-regime estimates with explicit, defensible weights (Section 3.1.3). Finally, convergence diagnostics across multiple chains (Appendix J) diagnose multimodality that would otherwise go unnoticed under a single EM run.

The Bayesian approach instead draws from the joint posterior:

$$p\big(\{s_t\}_{t=1}^{T},\ \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^{1},\ \mathbf{P} \,\big|\, \mathbf{z}_{1:T}\big)\,.$$

The states and the parameters are coupled: we can sample from the distribution of one block given the other, but not from either block on its own. Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, exploits exactly that structure by alternating between three conditional updates over the state path, the emission parameters, and the transition matrix. Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

Emission parameters

Before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. This is encoded by a conjugate Normal-Inverse-Wishart (NIW) prior on each regime's mean vector and covariance matrix:[5]

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \qquad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right). \tag{3}$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0}, \qquad \kappa_0 = 0.01, \qquad \nu_0 = D + 2, \qquad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1), \tag{4}$$

where D is the number of selected features. After observing data, these prior hyperparameters update to posterior counterparts κ_n, $\mathbf{m}_n$, ν_n, and $\boldsymbol{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. $\mathbf{m}_0 = \mathbf{0}$ is the natural prior

[5] The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix. Every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean.

given that the features are standardized. Without data, neither regime is presumed to center away from zero.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.

- ν_0 sets the Inverse-Wishart prior's degrees of freedom, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D+2$ is the smallest that yields a proper prior with finite mean; a larger choice tightens the prior around $\mathbf{I}_D$.

- $\mathbf{\Psi}_0$ is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\mathbf{\Sigma}_k] = \mathbf{\Psi}_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix

Regimes should be persistent rather than switching every few months. This is encoded by assigning each row of the transition matrix an independent Dirichlet prior whose mass concentrates on the diagonal:

$$\mathbf{P}_{0,\cdot} \sim \mathrm{Dir}(9,1), \qquad \mathbf{P}_{1,\cdot} \sim \mathrm{Dir}(1,9). \tag{5}$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\mathrm{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}.$$

This prior favors persistence in both regimes. Without data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}. \tag{6}$$

The prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. It is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

Initialization

The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude binary partition based on a single stress indicator is used:[6] months above

[6] The choice of initialization has negligible impact on the posterior. The sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 4.3, Appendix J) confirm that the chain mixes well regardless of the starting partition.

the indicator's median (the more stressed half) are initially assigned to the panic state ($s_t = 1$), and the rest to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \boldsymbol{\Sigma}_k^{(0)} = \mathrm{Cov}(\{\mathbf{z}_t : s_t = k\}),$$

where n_k is the number of months assigned to regime k. The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates

Each iteration cycles through three conditional updates: the latent state path $\{s_t\}_{t=1}^{T}$, the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, and the transition matrix $\mathbf{P}$. Each block is sampled from its full conditional distribution, with the other two blocks held fixed at their current values; for Steps 2 and 3 this distribution takes the conjugate Bayes form posterior $\propto$ likelihood $\times$ prior, while Step 1 (FFBS) samples the state path jointly from its full conditional via the forward-filtering backward-sampling recursion described below. The likelihood factor for Steps 1 and 2 is the Gaussian emission density $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, the plausibility of the observed feature vector $\mathbf{z}_t$ under regime k; Step 3 instead uses the multinomial likelihood of the sampled transition counts n_{ij}. The cycle is summarized graphically below; the steps that follow derive each update in turn.

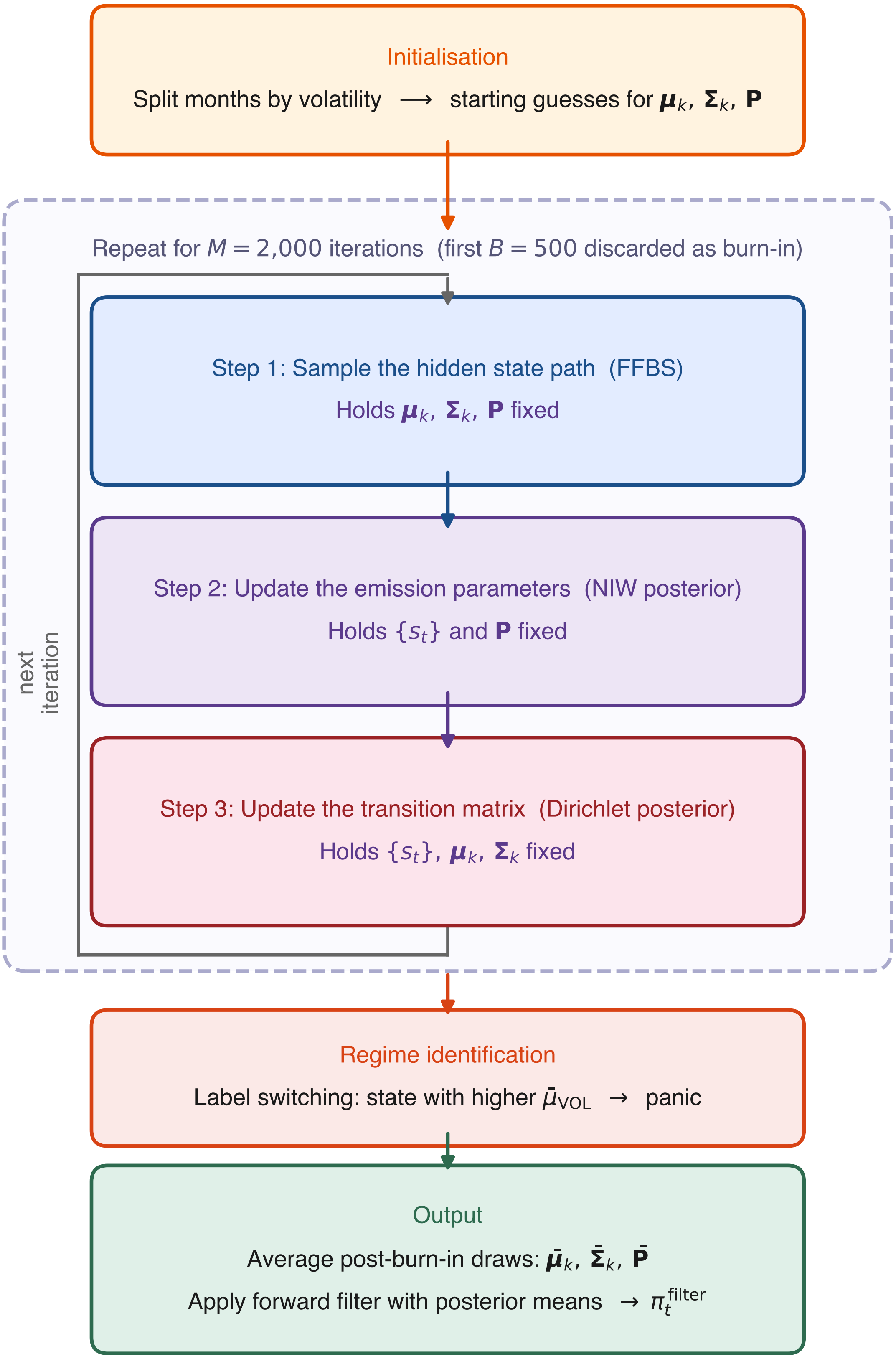

Figure 2: Flowchart of the Gibbs sampler.

Step 1: Sampling the hidden state path via FFBS

The full latent regime sequence $(s_1, \ldots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see Frühwirth-Schnatter, 2006, ch. 11), with the current emission parameters and transition matrix held fixed.

The forward pass (the *forward filter*) computes the filtered probability

$$\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t}),$$

the probability of being in state k at time t given all observations so far. $\alpha_t(k)$ is built in two sub-steps. First, a *prediction step* asks: before seeing this month's data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0)\, P_{0k} + \alpha_{t-1}(1)\, P_{1k}.$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes' rule. If the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k)\; \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0)\; \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1)\; \alpha_{t|t-1}(1)}, \tag{7}$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.[7] The recursion is initialized with a uniform initial state distribution $P(s_1 = k) = 1/2$, reflecting no prior preference for either regime. For the first observation this gives

$$\alpha_1(k) \propto f(\mathbf{z}_1 \mid s_1 = k).$$

The filtered probabilities $\alpha_t(k)$ are marginals. They describe each month's regime in isolation, not how consecutive months connect. Sampling each s_t independently from its α_t would ignore the transition matrix and could produce paths that flip regimes from one month to the next, in conflict with the strongly persistent $\mathbf{P}$. A second, backward pass is therefore needed to draw a full state path that respects $\mathbf{P}$. The backward pass proceeds in two steps:

[7] In practice, these densities can be extremely small (or large) for multi-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space. Log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalized.

Terminal state

At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \text{Cat}(\alpha_T(0),\, \alpha_T(1)),$$

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.[8]

Backward recursion

For each earlier time step $t = T-1, T-2, \ldots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1}, which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalizing gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k)\, P_{k,s_{t+1}}}{\alpha_t(0)\, P_{0,s_{t+1}} + \alpha_t(1)\, P_{1,s_{t+1}}}. \tag{8}$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the product $\alpha_t(0)\, P_{0,0}$ dwarfs $\alpha_t(1)\, P_{1,0}$, the conditional probability of $s_t = 0$ approaches one, and the algorithm very likely samples $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

The backward pass is used only during training. The sampled state path it produces feeds the conjugate Gibbs updates of the emission and transition parameters. The trading signal π_t^{filter} relies solely on the forward filter.[9]

Step 2: Updating the emission parameters

Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed:

$$\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \mathbf{S}_k = \sum_{t:\, s_t = k} (\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top.$$

[8] Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

[9] The forward filter uses only observations $\mathbf{z}_{1:t}$ (past and present) at each month t, never future months $\mathbf{z}_{t+1:T}$, so no look-ahead information leaks into the trading signal. The smoothed counterpart $P(s_t \mid \mathbf{z}_{1:T})$, which would condition on the full sample, is not used.

The posterior hyperparameters are:

$$\kappa_n = \kappa_0 + n_k, \qquad \mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n}, \qquad \nu_n = \nu_0 + n_k,$$
$$\boldsymbol{\Psi}_n = \boldsymbol{\Psi}_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top. \tag{9}$$

New regime parameters are then drawn from the updated posterior: $\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_n, \boldsymbol{\Psi}_n)$ and $\boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}(\mathbf{m}_n, \boldsymbol{\Sigma}_k/\kappa_n)$.

Step 3: Updating the transition matrix

The sampled path is scanned to count transitions, $n_{ij} = \sum_{t=2}^{T} \mathbb{1}(s_{t-1} = i,\, s_t = j)$, the number of times the path moved from state i to state j. Each row of the transition matrix is then drawn from its own Dirichlet posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \mathrm{Dir}\big(\alpha_{i0} + n_{i0},\ \alpha_{i1} + n_{i1}\big). \tag{10}$$

Sampling once from each of the two Dirichlet posteriors (one per regime) produces the updated transition matrix $\mathbf{P}$, which is held fixed during the next iteration's state-path sampling. Because the prior is strongly tilted toward persistence, the sampled $\mathbf{P}$ tracks the data while staying anchored around a persistent diagonal even on short or noisy paths.

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Seed Averaging

The Gibbs sampler is re-estimated on the expanding training window each year (Section 4.3). To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 50$ different random seeds. Each chain is run for $M = 2{,}000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. The forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_{t,s}^{\text{filter}}$. The final trading signal is the average across seeds:

$$\pi_t^{\text{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_{t,s}^{\text{filter}}. \tag{11}$$

Averaging across MCMC chains is a methodological requirement, not a performance optimization. A single Gibbs chain produces a noisy regime signal whose downstream Sharpe varies meaningfully across random initializations; averaging across chains reduces this noise, and the variance shrinks as more chains are added. By 50 chains the downstream Sharpe has effectively stabilized, with the plateau setting in well below that count (Appendix J.4). All out-of-sample numbers reported below are computed from this 50-chain ensemble, not from a single chain.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming

In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary. Relabeling the states leaves the likelihood unchanged, so the HMM identifies two clusters in the D-dimensional feature space without knowing which one should be called panic. To break this symmetry, each feature is assigned an expected sign during stress (the direction in which it moves during historical crisis windows relative to calm periods), and the cluster whose posterior mean vector best aligns with that pattern is designated as "panic" and the other as "calm". For the drawdown-only production model this reduces to designating the cluster with the deeper (more negative) mean drawdown as panic. The specific windows and sign assignments are listed in Appendix I. The HMM itself never sees these windows during training; they enter only at the post-estimation naming step.

The naming has no effect on the regime estimation itself. The two clusters and their boundaries are identical regardless of which is called calm or panic. What the naming controls is the interpretation of the subsequent analysis, which state we treat as the stress regime when reading off mechanism descriptions, conditional Sharpes, and cluster characterizations downstream. The label "panic" therefore reflects a statistical resemblance to historical crisis periods rather than a directly observed market state.

Filtered Regime Probability

Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t};\, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}). \tag{12}$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the cross-sectional model in Section 3.2.1.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into a cross-sectional portfolio. The production model is an XGBoost gradient-boosted tree regressor (XGB) that combines π_t^{filter} with momentum signals at twelve horizons to produce a monthly score for every stock. Each month, every stock in the point-in-time universe of the 1,000 largest by market capitalization is scored, and the top-scoring decile forms a long-only, value-weighted portfolio, held against the value-weighted top-1000 as the benchmark. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

One distinction from standard momentum strategies matters here. In the traditional approach (Jegadeesh and Titman, 1993), stocks are ranked by their raw trailing return at a fixed lookback, so the top decile always holds the highest-momentum stocks. Here, stocks are ranked by their predicted forward return, a learned function of the regime signal and the full momentum term structure, so the long portfolio need not contain high-momentum stocks at all. In panic, it can hold stocks below the cross-sectional average at every horizon (Section 5.2.2).

We evaluate the strategy in long-only construction, held against the value-weighted top-1000 benchmark, for two reasons. First, the deliverable is a large-cap, fully-invested mandate: the practitioner question is whether the regime signal improves an unlevered long book relative to the index it is measured against, not whether a dollar-neutral spread earns a premium. Second, restricting to the top-1000 and dropping the short leg removes the small-capitalization and short-side frictions that dominate long-short momentum results, isolating the regime-timing contribution on the tradable long side. Performance is therefore reported as active return, tracking error, and the information ratio against the benchmark rather than as a standalone Sharpe on a self-financing spread.

3.2.1 Cross-Sectional Stock Selection Model

An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Predicting the return rather than a binary above/below-median label preserves magnitude information: a stock expected to gain 10% ranks higher than one expected to gain 2%. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \ldots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal, for example by conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^{M} \eta \cdot h_m(\mathbf{x}), \tag{13}$$

where h_m is the m-th decision tree and η is the learning rate that shrinks each tree's contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}^s_{t+1} = F(\mathbf{x}^s_t)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 20 independent XGBoost models (each with a different random seed) is trained. Their predictions are averaged before ranking stocks into portfolio deciles.

Among the nonlinear architectures flagged in Section 2.4 (gradient-boosted trees, random forests, neural networks), XGBoost is preferred because its sequential residual fitting concentrates splits on whichever feature most reduces error, turning the regime signal into the model's primary gating variable rather than one feature among thirteen.

The maximum depth sets the order of feature interactions the model can represent. We fix it at four, the smallest order at which the regime signal can condition the joint configuration of several momentum horizons: depth 1 for the regime gate plus depths 2 to 4 for a three-horizon term-structure read, rather than an interaction with single horizons in isolation. Out-of-sample performance rises to a peak at depth 4 and declines for deeper trees, whereas training cross-validation cannot resolve depth: the validation Sharpe is flat within fold noise across depths, and although the single best-tuned configuration falls at depth 3, that lead is negligible against the noise (Appendix G.4). Depth 4 is adopted for that interaction-order reason; the out-of-sample peak corroborates it but, measured on the evaluation window, is not itself the selection criterion (Appendix G.4). This aligns the model's capacity with the mechanism examined in Section 5.2.2.

With the two-stage framework in place, the next section describes the data and sample construction, and develops the evidence, from the literature and from a train-only selection exercise, for market drawdown as the HMM's sole input feature.

4 Data

This section describes the stock-level sample and chooses the input for the regime classifier. The production HMM observes a single series: the market drawdown. Section 4.1 develops the case for conditioning on drawdown alone, grounded in the momentum-crash literature and confirmed by a selection rule restricted to training information; Section 4.2 defines the feature and explains what the HMM adds on top of it.

The stock-level sample comprises all ordinary common shares listed on the NYSE, AMEX, or NASDAQ with a price above \$1, from which a point-in-time universe of the 1,000 largest by month-end market capitalization is formed anew each month (a Russell 1000 proxy).[10] Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and −30% is imputed for performance-related delistings with missing returns.[11] Both the regime model and the cross-sectional model are estimated on an expanding window: for each trading year they use only data through the prior December, and the out-of-sample walk runs from January 2011 to November 2025 (179 months). Table 1 provides an overview.

Universe	Point-in-time 1,000 largest U.S. common stocks by month-end market cap (NYSE/AMEX/NASDAQ, price > \$1; Russell 1000 proxy)
Capitalization coverage	92.9% of listed U.S. market cap (monthly average)
Feature warm-up	Momentum lookbacks from January 1991
Initial training window	1991–2010 (240 months; 240,000 stock-months; 3,249 distinct stocks)
Out-of-sample walk	January 2011 – November 2025 (179 months; 179,000 stock-months; 2,312 distinct stocks), model re-estimated each January on an expanding window; historical extension back to 2000 in Section 5.3.3
Benchmark	Value-weighted portfolio of the same top-1000 universe

Table 1: Sample overview, applied configuration.

4.1 Choosing the Regime Indicator

The regime signal exists to answer a single question for the cross-sectional model: is the market currently in the state in which momentum's cross-sectional structure reorganizes? The choice of the HMM's observation series therefore determines which economic state the model can condition on. The production specification observes one series only: the market drawdown (DD), the percentage decline of the market index from its rolling 12-month peak, formally defined in Section 4.2. This subsection develops the case for conditioning on

[10] The \$1 floor follows Gu et al. (2020); value-weighting within the top-1000 keeps the realized portfolio in liquid, large-capitalization names.

[11] Full data sources and detailed variable definitions are provided in Appendix A.

drawdown alone. The momentum literature identifies the market's decline-and-recovery state, rather than market stress in general, as the conditioning information that matters for momentum, and drawdown is the most direct real-time measure of that state. Statistically, a univariate HMM extracts the state with the fewest estimated parameters. Finally, a mechanical rule that uses training information only, applied to a pool of drawdown-anchored candidates, adds no further stress feature to drawdown in any year of the evaluation window.

The crash literature points to the decline state

The momentum-crash literature ties the failure mode of momentum to one specific market condition: the aftermath of a large market decline. Daniel and Moskowitz (2016) document that momentum crashes are concentrated in months that follow bear markets, identified ex ante by a negative trailing two-year market return, and attribute the crash to the option-like payoff that the loser leg acquires after severe declines: past losers rebound violently when the market turns. Cooper et al. (2004) establish the same conditioning at lower frequency: mean momentum profits are positive following UP markets and vanish following DOWN markets, where the market state is the sign of the trailing three-year market return. Asem and Tian (2010) refine the market-state evidence along the same dimension: momentum profits are higher when the market continues in its prior state than when it transitions, and the transitions out of DOWN markets are where the strategy fares worst. Goulding et al. (2023) sharpen the state definition by intersecting slow and fast trailing market returns into four states (Bull, Correction, Bear, Rebound) and show that momentum's losses concentrate around market turning points, the exits from Bear states. The pattern also has a theoretical footing: in the equilibrium model of Cujean and Hasler (2017), disagreement rises as economic conditions deteriorate, return predictability concentrates in bad times, and momentum crashes after sharp rebounds out of those states. In each case the conditioning variable is a transformation of trailing market-level returns, the depth or direction of the recent decline, and not volatility, cross-sectional dispersion, or credit conditions. Drawdown is the member of this family that measures the decline state most directly and is observable in real time: it is zero at a market peak, deepens through a decline, and narrows as the recovery progresses.

Why not volatility, and why not more features

The leading alternative conditions momentum on second moments: Barroso and Santa-Clara (2015) scale the long-short portfolio by its own trailing realized volatility and eliminate the worst crashes, and Moreira and Muir (2017) generalize the recipe by scaling factor exposures by realized market variance. Volatility management, however, is a sizing rule: it decides how much of an unconditionally constructed portfolio to hold, while leaving the selection rule unchanged. The mechanism exploited here operates through selection: in the panic state the model rebuilds the long leg toward stocks positioned for recovery (Section 5), so the relevant conditioning information is the state itself, not the risk level. Volatility is elevated in any turbulent period, whereas drawdown is near zero except during and after declines, so drawdown isolates the decline-and-recovery state itself rather than turbulence in general. Volatility does carry conditional information for momentum: Wang and Xu (2015) show that high trailing market volatility forecasts low momentum payoffs, especially in down markets, and that volatility and the market state complement each other, and Daniel and Moskowitz (2016) find that the bear-market state and ex ante market variance forecast momentum returns independently. Two considerations still favor the univariate drawdown specification. The panic regime's wider emission variance lets the filter treat deep, fast-moving drawdowns as more panic-like, but this is a property of the drawdown series itself, not a substitute for a realized-return-volatility feature. The decisive evidence is empirical: the train-only selection rule described below, whose candidate pool includes drawdown-volatility combinations, never elects a volatility feature alongside drawdown.

A broader stress set faces a different objection: estimation cost without commensurate information. Each additional emission feature adds a regime-specific mean and its share of a regime-specific covariance matrix, so a two-state model on four features carries 30 free parameters against a training panel in which panic months are rare, while the univariate drawdown model carries six.[12] Rare-regime parameters estimated from few observations lean on the prior and are fragile out of sample. Indicators tied to particular market structures can also degrade silently when the setting changes; breadth- and participation-based measures, for instance, behave differently inside a large-cap universe than in the full cross-section. Parsimony also disciplines the specification search itself: the fewer modeling choices made on the researcher's side, the smaller the scope for selection bias in the reported results. Harvey et al. (2016) document how large the implicit search space behind published return predictors has become and argue that new findings must clear correspondingly higher statistical hurdles; fixing a single, economically motivated conditioning variable in advance keeps the regime layer's researcher degrees of freedom to a minimum.

A train-only rule adds no feature to drawdown

The literature-based choice is confirmed by a mechanical check that uses no evaluation-window information. A pool of 25 drawdown-anchored candidate combinations is built from eight monthly stress indicators, all observable in real time at month end: market drawdown (DD), realized volatility (VOL), and log VIX (LVIX) for equity-market stress; cross-sectional return dispersion (DISP), relative market participation (REL_N), and cross-sectional return skewness (SKEW) for cross-sectional structure; and the BAA–AAA

[12] For D features, each of the two states has D mean and $D(D+1)/2$ covariance parameters, and the transition matrix has two free entries: $2\,(D + D(D+1)/2) + 2$, which equals 6 for $D = 1$ and 30 for $D = 4$.

credit spread (CS) and the 10-year minus 2-year Treasury term spread (TERM) for credit and rates. The pool itself was fixed in advance using pre-2011 information only. Each candidate is evaluated by running the full two-stage pipeline on cross-validation folds whose validation windows end before the traded year, and candidates are compared on validation performance.[13] A one-standard-error rule then selects, among all combinations whose mean validation performance lies within one standard error of the best combination's, the one with the fewest features.[14] This rule elects the drawdown-only specification in every year of the 2011–2025 evaluation window. Because every candidate contains drawdown by construction, this exercise tests whether any additional stress feature earns its estimation cost on top of drawdown, not whether drawdown dominates every rival single indicator; the former is what the parsimony argument requires. The unrestricted alternative, taking the top-scoring combination each year, churns across six distinct multi-feature combinations over the same window without improving out-of-sample performance (Section 5). Selection is therefore not the engine of the results: the indicator is fixed on literature grounds, and the data-driven rule, restricted to training information, adds nothing to it.

4.2 The Drawdown Feature

Market drawdown is the percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \ldots, P_t)}{\max(P_{t-11}, \ldots, P_t)}. \tag{14}$$

DD_t equals zero when the market stands at a 12-month high and becomes increasingly negative as the market falls below its recent peak. Drawdown is a stock variable rather than a flow: it stays negative for as long as the market remains below its trailing 12-month peak, so it encodes not merely that a decline occurred but that the market has not yet climbed back above its recent high. It is therefore informative about the recovery phase, the window in which Daniel and Moskowitz (2016) locate the momentum crash and in which the selection mechanism studied in this paper activates; persistence of the panic classification beyond the trailing window is supplied not by the feature but by the HMM transition matrix.

The HMM contributes what a raw drawdown threshold would not. First, the filter output is a probability rather than a binary state: π_t^{filter} moves continuously as evidence accumulates, so the cross-sectional model can respond in proportion to regime uncertainty. Second, the transition matrix imposes persistence, so a single month of partial rebound does not immediately flip the state, whereas hard-threshold classifications built on trailing-

[13]The validation metric is the information ratio of the pipeline's portfolio against the value-weighted benchmark over the fold's validation months.

[14]The standard error is computed from the dispersion of the best combination's per-fold validation scores; ties at the minimum feature count are broken by the higher validation mean.

return signs, such as the UP/DOWN states of Cooper et al. (2004) or the ex ante bear indicator of Daniel and Moskowitz (2016), switch discretely at the boundary. Third, the level that separates the two states is learned from the training data rather than fixed in advance; Daniel et al. (2019), who model momentum's calm and turbulent states with exactly this class of two-state hidden Markov model (estimated on joint market and momentum returns), explicitly fault fixed-lookback definitions of depressed markets as arbitrary, and treating the boundary as a latent-state estimation problem answers that critique. The construction formalizes the point of Hamilton (1989) that the regime is latent and must be inferred probabilistically from what is observed.

The drawdown series is winsorized at the 1st and 99th percentiles[15] using training-sample statistics, and standardized to zero mean and unit variance.

Panel (a) of Figure 3 plots the standardized series over the sample. Panic regimes are characterized by deep drawdowns that are slow to close; calm regimes show shallow or no drawdown. The series deepens sharply in the 2000–2002 dot-com bust, the 2007–2009 Global Financial Crisis, the March 2020 COVID shock, and the 2022 tightening cycle, and sits near zero through the 2003–2007 and 2013–2019 expansions. The largest excursions align with the shaded NBER recession bands, though drawdown also flags equity-market stress that falls outside official recession dating.

4.3 Regime Estimation Results

The regime model is re-estimated every year of the walk rather than once on a fixed training block: for trading year Y the sampler runs on the drawdown panel through December $Y-1$, and the causal forward filter (Section 3.1.6) is applied through year Y. The regime signal evaluated below is therefore the concatenation of these out-of-sample yearly filters, the probability path that actually drove the portfolio, rather than a single in-sample fit.

Sampler configuration and convergence

The sampler settings are inherited from the validated companion long-short study: 2,000 Gibbs iterations with 500 discarded as burn-in, with the filtered probability averaged across independent seeds per fit (Section 3.1.5). Convergence of the companion study's multi-feature specification is documented at length in Appendix J via the effective sample size and the Gelman–Rubin $\hat{R}$; the drawdown-only model estimated here is a one-dimensional restriction of that specification, so its emission space is far better identified, and label switching, the only genuine multimodality in a two-state Gaussian HMM, is resolved deterministically by assigning the deeper-drawdown cluster to the panic state

[15] Winsorizing at the 1st and 99th percentiles caps any value below the 1st percentile at that bound and any value above the 99th percentile at that bound, limiting the influence of outliers without discarding the observations.

(Section 3.1.6). Averaging the filtered probability across seeds cancels the residual seed-specific Monte Carlo noise.

Regime separation and the filtered signal

Panel (b) of Figure 3 plots the resulting out-of-sample filtered probability π_t^{filter} over the 2011–2025 evaluation window. The signal is sharply bimodal: its median is 0.005 and it exceeds the 0.5 panic threshold in 45 of the 179 months (25%), so the great majority of months are classified as unambiguously calm. Elevated-probability episodes line up with the recognized stress windows of the period, the 2011 euro-area and U.S. downgrade sell-off, the 2015–2016 China and oil shock, the late-2018 drawdown, the March 2020 COVID crash, the 2022 tightening bear market, and the April 2025 tariff episode. The 2022 bear market is the longest single panic run at fourteen consecutive months.

The regime signal separates the states on drawdown exactly as intended. Across the evaluation window the market sits, on average, 9.9% below its trailing peak in panic-classified months versus 0.7% in calm months; in standardized units the mean drawdown is -0.57 in panic months against $+0.58$ in calm months. The filtered probability rises strongly with the depth of the drawdown, with a Pearson correlation of 0.86 between π_t^{filter} and the drawdown depth, but it responds to the full standardized-drawdown history rather than the contemporaneous level alone. The transition matrix supplies the persistence visible in the figure: isolated one-month dips do not trip the signal, whereas sustained declines push it toward one and hold it there until the drawdown closes.

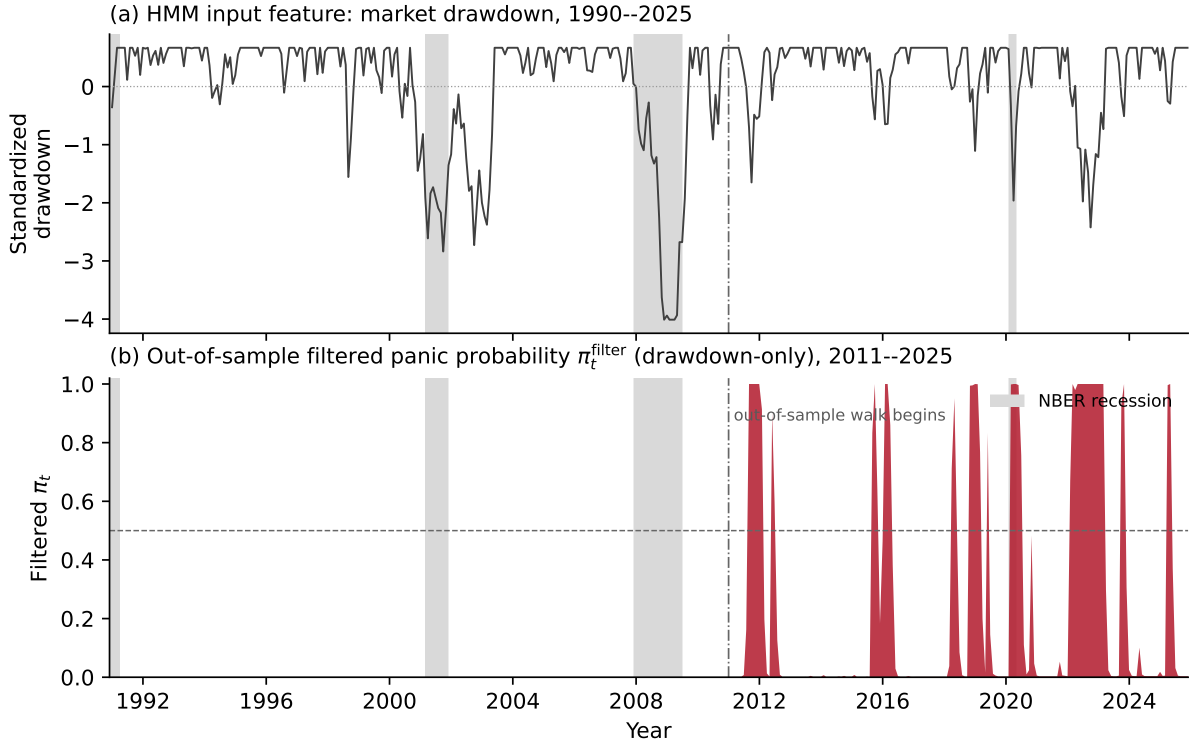

Figure 3: The drawdown regime signal. Panel (a): the standardized market drawdown, the sole HMM input feature, over 1990–2025; the dash-dotted line marks the start of the out-of-sample walk. Panel (b): the out-of-sample filtered panic probability π_t^{filter} from the drawdown-only model (the signal that drove the portfolio), 2011–2025, with the 0.5 panic threshold dashed. Grey bands mark NBER recessions.

4.4 Cross-Sectional Features

The cross-sectional model ranks stocks each month using twelve trailing momentum signals together with the filtered panic probability π_t^{filter}. The latter varies only at the monthly market level (every stock in the same month receives the same value), so it functions as a conditioning variable rather than a stock-level predictor.

Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\text{mom}_{k,t}^{s} = \prod_{\ell=1}^{k}(1 + r_{t-\ell}^{s}) - 1, \qquad k = 1, \ldots, 12, \tag{15}$$

where $r_{t-\ell}^{s}$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

4.5 International Sample

For the replication in Section 5.3.4, the same pipeline is applied to UK and Japanese equities from Compustat Global (monthly panels built from daily security files, total returns including dividends and splits, delisting metadata incorporated; data through mid-2026). No per-region tuning is performed: the regime signal is the same univariate drawdown HMM estimated from each region's own market panel, and the large-cap universe is chosen mechanically as the smallest top-N covering 90% of listed market capitalization on the pre-2011 sample (UK $N = 200$, JP $N = 750$), mirroring the U.S. top-1000 construction. Portfolios are long-only top-decile and value-weighted within the regional universe, measured against the regional value-weighted benchmark over the matched 2011–2025 window.

With the regime signal and the cross-sectional features in place, the next section evaluates the regime-conditioned portfolio out-of-sample and decomposes the model's monthly selections.

5 Results

This section evaluates the drawdown-conditioned momentum strategy out-of-sample on the applied configuration: a long-only top-decile portfolio, value-weighted, drawn from the point-in-time top-1000 universe, rebalanced on an expanding yearly walk over 2011–2025, and measured against the value-weighted top-1000 benchmark. Section 5.1 reports headline performance and shows that the strategy earns no unconditional risk-adjusted alpha but concentrates its entire edge in panic regimes. Section 5.2 decomposes the model's per-month selections and locates that edge in a deep-crisis loser-selection mode. Section 5.3 stress-tests the result across episodes, a risk-aversion sweep, a pre-2011 historical backtest, and an international replication. Section 5.4 takes up the transaction-cost case for the regime signal: that it relocates trading to the months where it is worth paying for.

5.1 Portfolio Performance

5.1.1 Results Overview

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	IR	Final \$
VW top-1000 benchmark	14.1%	14.4%	0.99	-24.3%	1.00	–	7.2
Unconditional 12-1 mom	15.2%	20.7%	0.79	-30.6%	1.15	0.16	8.3
XGB, no regime signal	14.2%	22.3%	0.71	-43.1%	1.30	0.12	7.3
XGB + π (argmax)	12.3%	21.7%	0.64	-38.0%	1.27	-0.03	5.7
XGB + π (DD, main)	15.6%	21.6%	0.78	-32.4%	1.28	0.21	8.7

Table 2: Out-of-sample portfolio performance, applied configuration.

Notes: Long-only top-decile, value-weighted, point-in-time top-1000 universe, *gross* of costs (transaction costs are treated separately in Section 5.4). β is the full-sample market beta; IR is the information ratio of monthly active returns against the value-weighted top-1000 benchmark. Test period: January 2011 to November 2025 (179 months). The main model conditions on the drawdown-only regime probability π_t^{filter} selected by the parsimonious rule (Section 4.1); the argmax row conditions on the yearly best-scoring feature combination; the no-signal row drops π_t^{filter}.

Table 2 reports gross out-of-sample performance for the main model, the drawdown-conditioned XGBoost regressor, alongside its no-signal ablation, an alternative feature-selection variant, unconditional 12-month momentum, and the value-weighted top-1000 benchmark. The honest headline is stated first: on a large-cap, long-only book the strategy does *not* deliver risk-adjusted outperformance. Its Sharpe ratio of 0.78 is below the benchmark's 0.99, and although it compounds \$1 into \$8.7 against the benchmark's \$7.2 over the window, it does so at higher volatility (21.6% vs 14.4%) and higher market beta (1.28). The excess return is a beta-and-tilt effect, not alpha: the six-factor alpha is an insignificant +1.6% per year ($t = 0.34$; Appendix H.3), and even the CAPM alpha (+2.8%,

$t = 0.55$) does not clear conventional thresholds.[16] This is a sharp departure from the long-short, full-universe setting, where the same architecture earns a large and significant alpha: stripping the short leg, restricting to large capitalizations, and measuring against the index the book is actually held against removes the headline result.

What survives is a timing result. The information ratio against the benchmark is 0.21 for the main model, the highest of the conditional variants, and positive where dropping the regime signal (0.12), feeding the raw stress indicators to the tree (0.12), or selecting the yearly best-scoring feature combination (−0.03) are weaker. The regime signal adds value, but that value is an active-return, benchmark-relative quantity, not a standalone premium.

The regime-conditional view (Table 3) shows where it comes from. Panic is defined by the main model's own filtered probability at formation ($\pi_t^{\text{filter}} \geq 0.5$), the same classification as Section 4.3: 45 panic and 134 calm months.[17] In calm months the strategy is essentially the index with extra volatility: its mean active return is +0.04% per month, and its calm Sharpe (0.75) trails the benchmark's (1.10) only because it carries the same return at higher risk. In panic months the active return is +0.72% per month (roughly 9% annualized), with the strategy matching the benchmark's panic Sharpe (0.90) despite its higher volatility. Effectively all of the strategy's active return is earned in the 45 panic months; the calm months contribute nothing but also cost nothing. The practitioner reading is direct: be the index in calm, tilt in panic.

The comparison across rows in Table 3 isolates what the regime signal contributes. The no-signal variant earns *more* panic-month active return (+0.82% per month) than the main model, but it bleeds −0.11% per month through the 134 calm months; unconditional 12-1 momentum shows the same pattern in weaker form. The drawdown signal's contribution is therefore not to amplify the panic-month payoff, which is available to any momentum-tilted book, but to keep the portfolio benchmark-like through the long calm stretches in which the tilt otherwise loses money. That discipline is what lifts the full-window information ratio to 0.21 against 0.12 without the signal, and it is the observation on which the transaction-cost case of Section 5.4 is built.

[16] The factor alpha is the intercept of a regression of monthly excess returns on the market, SMB, HML, RMW, CMA, and UMD factors; full specification in Appendix H.3.

[17] An earlier reporting convention classified months with the argmax selection variant's probability (54 panic months), under which the regime contrast appears sharper (panic Sharpe 1.35, calm mean active −0.28% per month for the main model). All regime-conditional numbers in this section use the main model's own signal; the qualitative conclusion, that the active return is concentrated in panic months, holds under either classification.

	Sharpe			Mean active (%/mo)	
	Full	Calm	Panic	Calm	Panic
VW top-1000 benchmark	0.99	1.10	0.90	–	–
Unconditional 12-1 mom	0.79	0.67	1.09	−0.01	+0.71
XGB, no regime signal	0.71	0.59	0.99	−0.11	+0.82
XGB + π (argmax)	0.64	0.55	0.87	−0.24	+0.59
XGB + π (DD, main)	0.78	0.75	0.90	+0.04	+0.72

Table 3: Regime-conditional performance, applied configuration. Panic is defined by the main model's own filtered probability at formation ($\pi_t^{\text{filter}} \geq 0.5$): 45 panic and 134 calm months, the same classification as Section 4.3. Mean active is the average monthly return over the value-weighted top-1000 benchmark within the regime.

The panic edge does not convert into factor alpha, and this qualification must travel with the result. Estimated on the 45 panic months alone, the main model's CAPM alpha is +10.6% annualized ($t = 0.98$) and its six-factor alpha only +3.3% ($t = 0.32$); the calm-subsample six-factor alpha is +1.7% ($t = 0.34$). The benchmark itself earns a panic-month CAPM alpha of +5.1% ($t = 1.18$), so part of what the strategy harvests in panic is a market-wide recovery effect that the static factors do not span, leveraged through its higher beta, rather than a cleanly separated selection premium.[18] The defensible claim is therefore regime-conditional active return, together with the transaction-cost case that Section 5.4 develops, not risk-adjusted alpha.

[18] The subsample alphas are sensitive to the regime classification: under the earlier common classification the panic six-factor alpha is +16% ($t = 1.15$) on 53 months. No classification produces a statistically significant panic alpha, which is why the section's claims rest on active returns rather than subsample alphas.

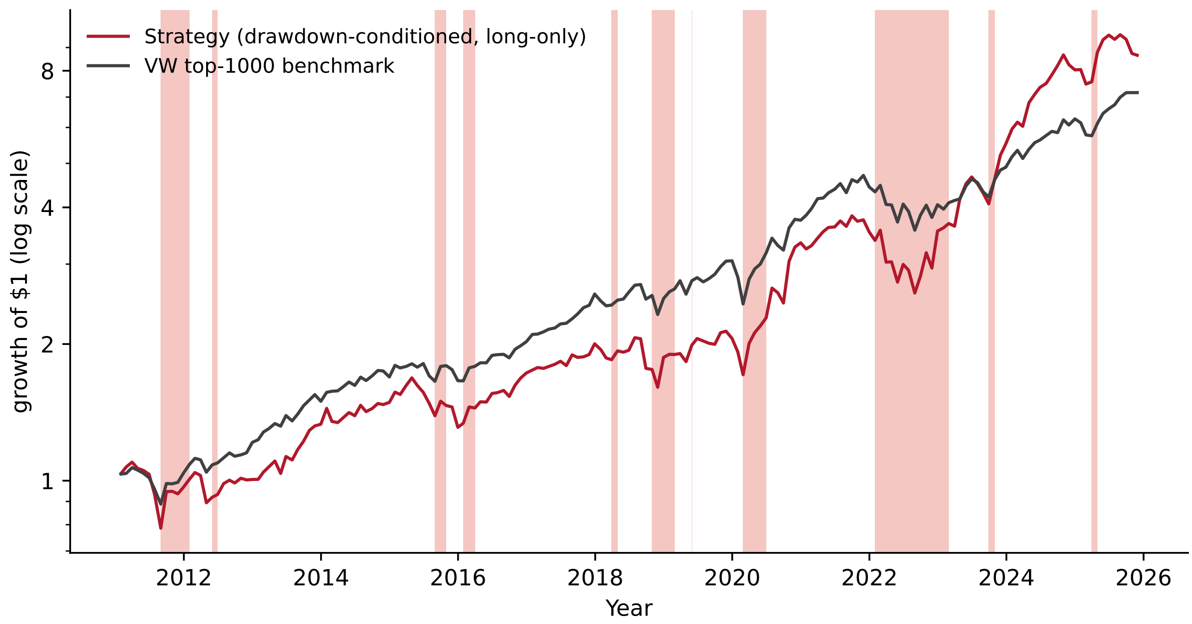

Figure 4: Cumulative wealth paths (log scale) from $1 in January 2011: the main model against the value-weighted top-1000 benchmark. Shaded bands indicate panic months under the main model's own signal ($\pi_t^{\text{filter}} \geq 0.5$; 45 months).

The Newey–West truncation of six lags is supported by Ljung–Box diagnostics on the six-factor residuals of the main model: LB(6) $= 4.98$ ($p = 0.55$) and LB(12) $= 7.62$ ($p = 0.81$), so the null of no residual autocorrelation is not rejected.

5.1.2 Regime Signal Ablation

The regime signal is what does the work in point estimate, and the HMM stage is what makes it usable (Table 4). Dropping π_t^{filter} and retraining on the twelve momentum features alone lowers the information ratio from 0.21 to 0.12. Feeding four raw stress indicators (DD, VOL, DISP, REL_N) to the tree directly, rather than the HMM-distilled probability, does no better (0.12): with the extra dimensions the model must reconstruct the regime structure at each split, whereas the HMM solves regime identification upstream and hands the cross-sectional model a single conditioning variable one split can act on. The GHM turning-point rule, evaluated at the seed budget matching production, also trails (0.10), and adding firm-level fundamentals degrades the strategy (-0.12; Section 5.1.3). With 179 months, the IR differences across variants are directional evidence rather than individually significant, but they point the same way throughout: the drawdown-only HMM signal is the operative feature, and its parsimony (Section 4.1) is a feature rather than a limitation.

Regime Signal Variant	Sharpe	IR
XGB + π_t^{filter} (HMM, drawdown; main)	0.78	0.21
XGB, no regime signal	0.71	0.12
XGB + four raw indicators (DD, VOL, DISP, REL_N)	0.72	0.12
XGB + GHM turning-point rule	0.71	0.10
XGB + 10 fundamentals	0.69	-0.12

Table 4: Regime-signal ablation, applied configuration (IR against the value-weighted top-1000 benchmark). The HMM-distilled drawdown probability π_t^{filter} improves on dropping the signal and on feeding the raw stress indicators to the tree directly. Seed budgets: the main and no-signal rows use the production 20-seed budget; the raw-indicator and fundamentals rows use the 5-seed ablation budget; the GHM row is reported at the full 20-seed budget matching production (at the 5-seed budget its IR is 0.27, so the estimate is seed-sensitive and the matched-budget figure is the fair comparison). The fundamentals row covers 2011–2024 (fundamentals unavailable for 2025). REL_N-only is omitted: the single-feature HMM degenerates to $\pi \equiv 1$ post-2000 in the large-cap universe (a constant, hence equal to no-signal).

5.1.3 Fundamental Feature Augmentation

A complementary question is whether *adding* firm-level fundamentals to the feature set strengthens the strategy. Ten characteristics are tested,[19] each trained at production settings.

	Feat.	Sharpe	IR	Calm Sh.	Panic Sh.	Panic act. (%/mo)
Baseline (mom + π; main)	13	0.78	+0.21	0.75	0.90	+0.72
Baseline + 10 fundamentals	23	0.69	−0.12	0.68	0.72	−0.03
Fundamentals only	10	0.76	+0.07	0.73	0.88	+0.45
Momentum + fund. (no π)	22	0.77	+0.20	0.65	1.05	+0.85

Table 5: Feature-set ablation, applied configuration (5-seed ablation budget; regime split on the main model's own π). IR and active returns are measured against the common value-weighted top-1000 benchmark for every row. The fundamentals-augmented baseline covers 2011–2024 (fundamentals unavailable for 2025); other rows cover the full 179 months.

The results split cleanly along the regime mechanism (Table 5). Adding the ten fundamentals *on top of* the regime-conditioned baseline is destructive: the information ratio falls from +0.21 to −0.12, and the loss is concentrated exactly where the strategy earns its keep, with panic-month active return collapsing from +0.72% to −0.03% per month. Fundamentals describe firm-level characteristics whose cross-sectional ranking does not flip with the regime the way the optimal momentum tilt does, so inside the regime-conditioned model they dilute the panic-side selection that carries the active return. A

[19] Book-to-market, return on equity, gross profitability, asset growth, leverage, earnings growth, log market equity, operating cash flow over assets, free cash flow over assets, and accruals; definitions in Appendix L.

fundamentals-only model is close to the benchmark (IR +0.07). The interesting row is momentum plus fundamentals *without* the regime signal: it reaches a comparable full-window information ratio (+0.20) by a different route, a stronger panic tilt (+0.85% per month) bought with a weaker calm profile (calm Sharpe 0.65 against the main model's 0.75). With 179 months these IR differences are directional rather than individually significant; what the panel establishes is that fundamentals do not combine with the regime signal, not that they carry no information. A regime-conditional design that routes fundamentals into calm-regime predictions and momentum into panic-regime predictions is discussed in Section 6.3. Fund-level SHAP shares are not recomputed for the applied configuration.

5.2 The Regime-Momentum Mechanism

This section reads the structure of the regime-momentum signal off the model's per-month selections: first the aggregate contribution by feature (Section 5.2.1), then the recurring selection modes recovered by clustering the long-leg z-curves (Section 5.2.2), and finally the interpretation (Section 5.2.3).

5.2.1 Momentum Contribution by Horizon

The model's predictions are decomposed using SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017), which attribute the prediction across inputs (Appendix D.1), computed from walk-consistent yearly retrains over the out-of-sample months. A single feature, π_t^{filter}, carries 37% of total SHAP, with the twelve momentum horizons together accounting for the other 63% (Table 6). Equal weighting would give π_t^{filter} just $1/13 \approx 8\%$, so the model loads on the regime signal roughly five times its share of the feature set, consistent with a routing-variable role: the regime probability decides how the momentum term structure is used, while the momentum features do the stock-by-stock ranking. The split across regimes supports this division of labor. The regime signal's share is somewhat *lower* in panic (33%) than in calm (38%): once π_t^{filter} saturates near one and the model has routed into its reversal branch, the term structure carries the differentiation across stocks. The same gradient appears within the long leg, where the momentum features carry 75% of the attribution for the stocks the model actually selects.

The per-horizon breakdown (Figure 5) concentrates on the long lookbacks: months 11 and 12 carry the two largest shares of momentum attribution (22% and 18%), consistent with the classical Jegadeesh and Titman (1993) twelve-month momentum range, with a secondary role for the intermediate month-4 and month-8 horizons and little weight on the shortest lookbacks. The profile is nearly identical for the long-leg picks, so the model reads the same part of the term structure whether it is scoring the universe or the stocks it ends up holding.

Feature	Overall	Calm	Panic	Long-leg picks
Momentum (12 horizons)	63%	62%	67%	75%
π^{filter}	37%	38%	33%	25%

Table 6: SHAP feature-importance shares for the main XGB model, applied configuration (walk-consistent yearly retrains, three seeds; shares of total |SHAP| over the out-of-sample stock-months, with the regime split on the main model's own π). Each column sums to 100%.

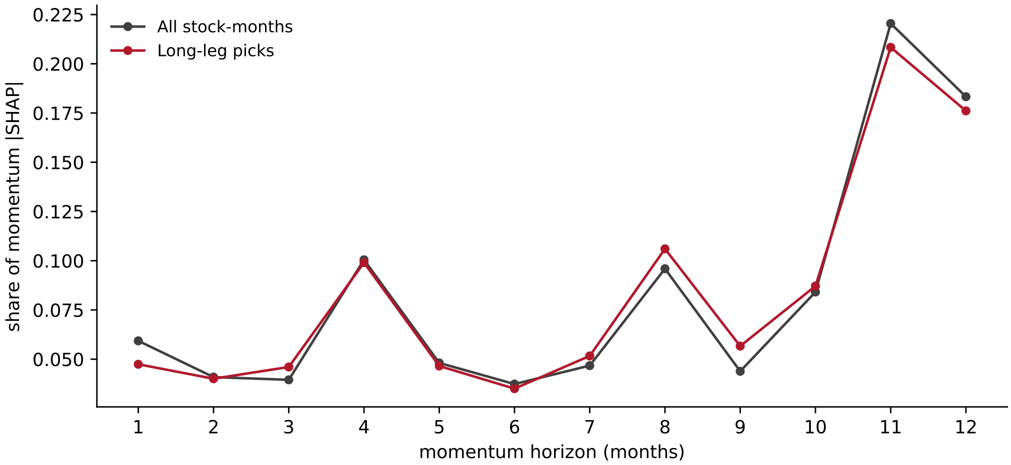

Figure 5: Per-horizon shares of momentum |SHAP| in the main model's predictions, applied configuration: all out-of-sample stock-months versus the long-leg picks.

5.2.2 Cross-Sectional Z-Scores

To trace the picks themselves, we examine the cross-sectional z-scores of the stocks the model selects for the long leg. For each selected stock s the z-score at horizon k is $(mom_k^s - m\bar{o}m_k)/\sigma_k$, where $m\bar{o}m_k$ and σ_k are the cross-sectional mean and standard deviation of horizon-k trailing momentum across the universe that month. The long-leg z-curve for a month is the average of these per-stock z-scores, a 12-element profile locating the model's picks relative to the cross-sectional mean at each lookback. The sign indicates which side of the mean the basket sits on; the magnitude, how far, in standard-deviation units.

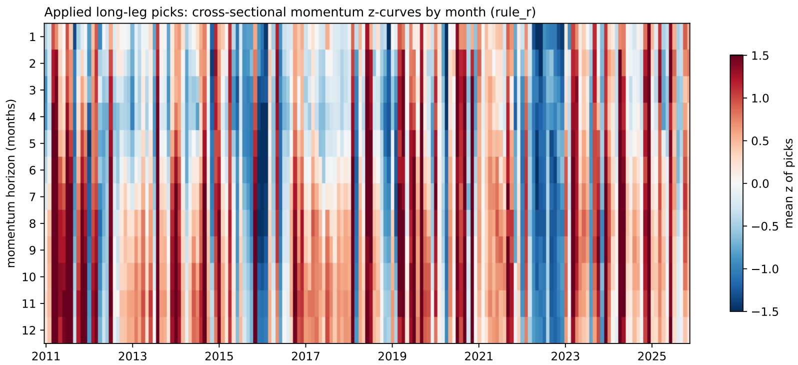

Figure 6: Long-leg cross-sectional z-scores by month and momentum horizon, 2011–2025. Each row is one of the 179 test months sorted chronologically; each column is one of the 12 momentum horizons; color shows the deviation of the long leg's mean momentum at that horizon from the cross-sectional mean that month.

Clustering the 179 monthly z-curves with K-means (L_2, $K = 4$) on the applied strategy's own picks recovers four recurring selection modes (Table 7).[20] The modes are ordered by the tilt of the picks, not by regime probability. Clusters 1 and 2 sit above the cross-sectional mean (average z-scores +1.23 and +0.29): classical winner-continuation, strongly tilted in cluster 1 and nearly benchmark-like in cluster 2, which is also the calmest group (mean $\pi = 0.16$). Clusters 3 and 4 sit below the mean (−0.33 and −1.01): the long leg is rebuilt from cross-sectional losers, mildly in cluster 3 and emphatically in cluster 4. Mean regime probability cleanly separates only the deep-crisis cluster (0.70); clusters 1 and 3 sit at similar moderate levels (0.28 and 0.33), so the loser tilt is triggered by the joint configuration of the regime signal and the term structure rather than by the probability level alone.

	1 (continuation)	2 (benchmark-like)	3 (mild reversal)	4 (deep crisis)
Months (n)	49	70	33	27
$\bar{\pi}_t^{\text{filter}}$ (mean panic prob.)	0.28	0.16	0.33	0.70
Long-leg avg z-score $\bar{z}$	+1.23	+0.29	−0.33	−1.01
Cluster Sharpe	0.76	0.84	0.66	0.92
Cluster IR (vs benchmark)	+0.33	+0.03	−0.30	+0.73

Table 7: $K = 4$ cluster descriptors, applied configuration, 2011–2025 ($n = 179$ months). Clusters are formed by K-means (L_2) on the 12-dimensional monthly long-leg z-curves of the main model's own picks and labeled by their z-tilt: only the deep-crisis cluster is clearly elevated in regime probability, while clusters 1–3 sit at moderate mean π (the benchmark-like cluster is in fact the calmest). $\bar{z}$ is the average cross-sectional z-score of the long leg across horizons (positive: picks above the cross-sectional momentum mean; negative: below). Cluster Sharpe and IR are computed over the months assigned to each cluster.

[20] $K = 4$ is retained for comparability with the mechanism decomposition; the clustering is run on this configuration's own long-leg z-curves rather than transplanted from the long-short, full-universe model.

The performance across modes is the central result of the decomposition. The deep-crisis mode (cluster 4) earns the highest cluster Sharpe, 0.92, and by far the highest information ratio, +0.73: when the model commits to deep-loser selection at the panic peak, the bet pays. The strong-continuation mode (cluster 1) earns a Sharpe of 0.76 and a positive IR (+0.33), the benchmark-like mode (cluster 2) tracks the index (0.84, IR +0.03), and the mild-reversal mode (cluster 3) is the weakest, with a Sharpe of 0.66 and a *negative* information ratio (−0.30). The edge is thus concentrated in the deepest-stress months and in confident continuation; the half-committed reversal months detract. No single-horizon rank-based momentum strategy would select the cluster 3 and 4 baskets for the long side: both centroid z-curves are negative at every one of the twelve horizons, and cluster 4 months individually never place the long leg above the cross-sectional mean at any horizon (in 17 of the 33 cluster 3 months at least one horizon edges above the mean, so the statement holds at the centroid level and exactly for the deep-crisis mode).

5.2.3 Discussion

The mechanism is therefore "momentum where momentum works, reversal where momentum crashes": above-average winner selection in calm (clusters 1–2) and below-average loser selection in panic (clusters 3–4), with the strongest active return coming from the deep-crisis loser bet. Because the model ranks on predicted forward return rather than on a per-horizon momentum score, the long leg can hold stocks not top-ranked at any individual horizon, which is exactly what implements the regime-conditioned direction flip. This is the same qualitative mechanism documented in the long-short, full-universe setting; what changes in the large-cap long-only setting is the *magnitude*: with the short leg gone and the universe restricted to liquid large caps, the reversal bet still works but its Sharpe is much smaller, and the strategy no longer beats a passive index on a risk-adjusted basis.

The economic reading connects to the momentum-crash literature. In panic, past losers among large caps are disproportionately high-beta names that fell hardest; conditioning on π_t^{filter} places them in the long leg, so the same rebound that crashes unconditional momentum (Daniel and Moskowitz, 2016) lifts the strategy instead. The premise is that panic-period declines are temporary; where it fails, in a prolonged deterioration, the reversal bet bleeds, which the historical backtest in Section 5.3.3 examines directly.

5.3 Robustness and Limitations

5.3.1 Episode Evidence

Three out-of-sample stress episodes fall inside the window (formation-month cumulative returns). Through the COVID crash and recovery the main model returned +62.4%

against the benchmark's +22.8%; through the April 2025 tariff episode, +28.2% against +14.2%; through the 2022 tightening bear market, +0.5% against the benchmark's −8.6%. In each the strategy outperformed the index, consistent with the deep-crisis selection mode of Section 5.2.2; direct evidence on the hold-versus-liquidate margin is part of the execution analysis in Section 5.4. Episode outperformance is not unique to the regime-conditioned model, however: in the COVID episode the no-signal variant (+89.3%) and unconditional momentum (+82.0%) gained even more, reinforcing that the regime signal's distinct contribution is calm-month discipline rather than a larger panic payoff.

5.3.2 Risk-Return Trade-off

The strategy's volatility (21.6%) is high relative to the benchmark's (14.4%) and a risk-averse investor may prefer to scale it down. A risk-aversion parameter γ is incorporated at the stock level: each stock is scored by $\operatorname{sign}(\hat{r}) \log(1 + |\hat{r}|/\varepsilon) - \gamma \log \hat{\sigma}$, where $\hat{r}$ is the model's return prediction and $\hat{\sigma}$ is a forward twelve-month volatility forecast from a second XGBoost model trained alongside the walk (ten seeds, trailing-volatility feature added; construction follows Appendix F), and the long portfolio takes the top utility decile. At $\gamma = 0$ the score is a monotone transform of $\hat{r}$ and reproduces the headline strategy exactly (Sharpe 0.78, volatility 21.6%), which validates the construction.

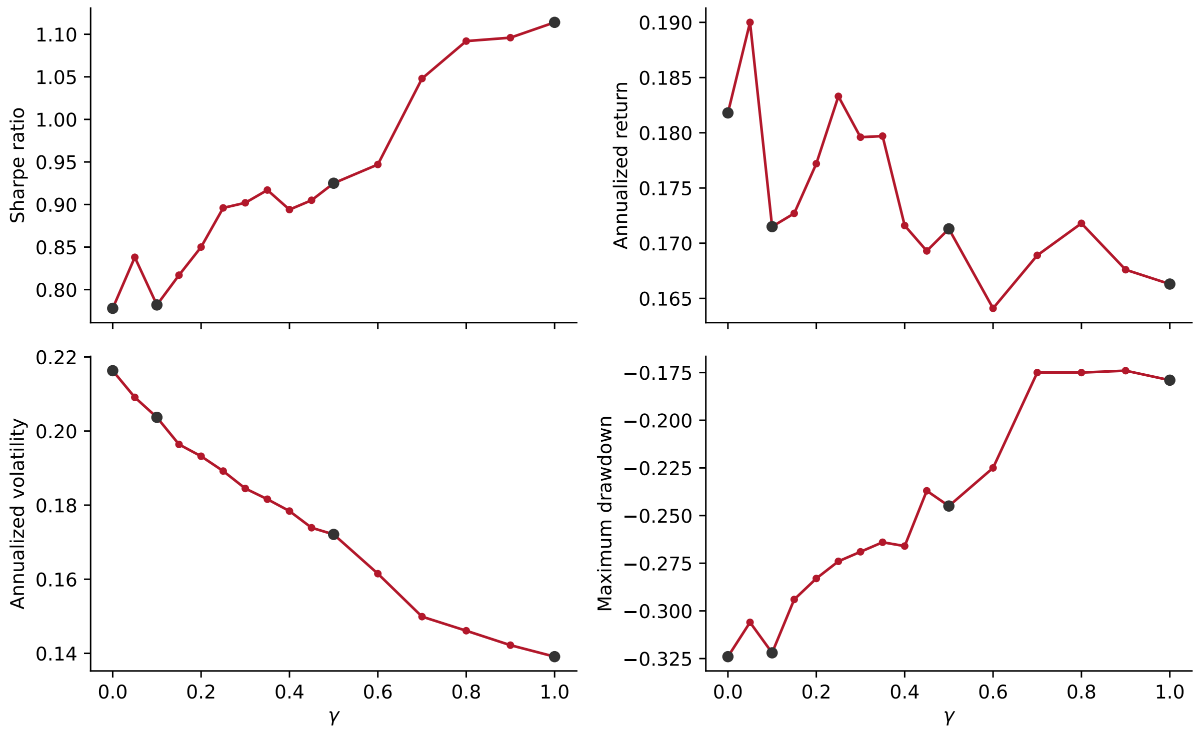

Figure 7: Risk-aversion sweep, applied configuration: Sharpe ratio, annualized return, annualized volatility, and maximum drawdown as a function of γ over $[0, 1]$. Marked points: $\gamma \in \{0, 0.1, 0.5, 1.0\}$.

The sweep delivers a different, and for a practitioner friendlier, conclusion than the long-short setting. There, the volatility penalty progressively removed the volatile beaten-down names that carry the panic reversal, trading Sharpe away for risk reduction. Here the trade-off largely disappears: as γ rises to one, volatility falls from 21.6% to 13.9% and the maximum drawdown from −32% to −18%, while the Sharpe ratio *improves* from 0.78 to 1.11 and the information ratio fluctuates in a 0.12–0.29 band with no systematic decay (0.15 at $\gamma = 1$). In a long-only large-cap book the volatility penalty tilts the portfolio toward the low-volatility segment of the universe, which historically earns close to the market return at lower risk, and this tilt stacks with, rather than cancels, the regime-timing active return. The risk control is therefore approximately free in this configuration: an investor can run the strategy at benchmark-like volatility (13.9% against the benchmark's 14.4%) without giving up the panic-month edge. These are single-path point estimates without inference, so we read the sweep as establishing feasibility, not as an optimized configuration.

5.3.3 Historical Stress: Out-of-Sample Evidence

The 2011–2025 window excludes the worst momentum-relevant stress: the dot-com bust, the global financial crisis, and the 2009 momentum crash. To extend the evaluation, the full pipeline is re-estimated on the same annual expanding-window schedule back to 2000, with each year's model trained only on data through the prior December and the drawdown feature standardized point-in-time within each fold, so no look-ahead enters the extension. Table 8 reports the stitched 2000–2025 series (311 months) by sub-period.

		Strategy			Benchmark	
Period	N months	Sharpe	Total Return	Max DD	Sharpe	Total Return
Full sample, 2000–2025	311	0.44	757.2%	−77.3%	0.62	743.0%
2000–2002 dot-com	36	−0.48	−65.6%	−77.3%	−0.70	−36.4%
2003–2006 bull	48	1.50	140.7%	−11.0%	1.89	83.6%
2007–2009 GFC	36	−0.03	−17.1%	−59.3%	−0.24	−18.3%
2009 momentum crash (Mar–Dec)	10	1.81	49.8%	−6.8%	2.72	36.2%
2010–2019 calm	120	0.69	197.4%	−28.5%	1.12	276.3%
2020–2025 COVID era	71	1.06	320.5%	−32.4%	0.91	134.7%

Table 8: Sub-period decomposition of the extended expanding-window backtest, 2000–2025, applied configuration (long-only top-decile, value-weighted, point-in-time top-1000, gross). Years 2000–2010 are generated by the same yearly expanding walk with the drawdown-only specification imposed directly and point-in-time drawdown standardization per fold; 2011–2025 is the production walk. Benchmark is the value-weighted top-1000.

The extension delivers one warning and two confirmations. The warning is the dot-com bust: over 2000–2002 the strategy lost 65.6% against the benchmark's 36.4%, with the full-sample maximum drawdown of 77.3% set in this episode. This is the reversal-failure mode realized: the regime signal correctly classified the period as panic (the filtered probability sat near one through 2001–2002), but in a prolonged deterioration the beaten-down stocks the model rotates into kept falling because their businesses were failing, and a fully-invested long book has no exposure lever to pull. Any deployment of the strategy must treat this as its worst-case scenario, and it is the strongest argument for the banded, cost-aware implementation of Section 5.4 rather than a mechanical full-conviction one. The first confirmation is the 2009 momentum crash, the episode that defines the crash literature (Daniel and Moskowitz, 2016): over March–December 2009 the strategy gained 49.8% against the benchmark's 36.2%, so the window that destroys unconditional momentum leaves this strategy ahead of its index. The second is the regime pattern itself: the strategy trails the benchmark through the calm 2003–2006 and 2010–2019 stretches and leads through the stressed 2020–2025 era, the same calm-drag, panic-edge signature as the production window. Over the full 25 years the strategy's Sharpe (0.44) trails the benchmark's (0.62) with comparable cumulative return, reinforcing that the unconditional case for the strategy is weak and the conditional, cost-aware case is the right one.

5.3.4 International Robustness

The pipeline is replicated on UK and Japanese equities (Compustat Global, data through mid-2026), with no per-region tuning: the same drawdown-only regime signal estimated from each region's market panel, the same long-only top-decile construction inside a mechanically chosen large-cap universe, and the same expanding walk over the matched 2011–2025 window (Table 9).

	UK (top 200)				JP (top 750)			
Strategy	Ann. Ret	Sharpe	Max DD	IR	Ann. Ret	Sharpe	Max DD	IR
Local VW benchmark	7.5%	0.64	−23.5%	–	12.9%	0.83	−22.0%	–
Unconditional 12-1 mom	13.4%	0.79	−26.3%	+0.43	12.3%	0.60	−29.3%	−0.04
Regime-conditioned (main)	10.2%	0.50	−30.8%	+0.19	15.2%	0.76	−33.3%	+0.22

Table 9: International replication, applied configuration, 2011–2025 (179 months, matched to the US window). Long-only top-decile value-weighted portfolios within mechanically chosen large-cap universes (smallest top-N covering 90% of listed market capitalization on the pre-2011 sample: UK $N = 200$, JP $N = 750$); drawdown-only regime signal from the regional market panel; same expanding walk and frozen hyperparameters as the US pipeline, no per-region tuning. IR is against the regional value-weighted benchmark.

Each region replicates a different component of the US result, and neither replicates both. Japan replicates the headline value proposition. Unconditional 12-month momentum is dead there, as it has been throughout the literature (Asness et al., 2013): the 12-1 portfolio earns an information ratio of −0.04 against the local benchmark. The regime-conditioned model turns the same inputs into +0.22, so where the unconditional signal has nothing to repackage, the regime conditioning still manufactures active return. The UK replicates the concentration mechanism instead. There, plain momentum works (+0.43) and the regime-conditioned model trails it (+0.19); but the model's active return is panic-concentrated exactly as in the US (+0.58% per month in the 50 panic months against +0.07% in calm), whereas plain UK momentum earns its keep in calm months and *loses* in panic (+0.69% calm, −0.19% panic). The two strategies are therefore complements in the UK rather than substitutes, harvesting different states.

The honest summary is that the mechanism travels but the full US configuration does not: regime-conditioned selection adds value where momentum fails (Japan) and reshapes when the value arrives where momentum works (UK), while maximum drawdowns run deeper than the regional benchmarks in both markets, as the strategy's high-beta panic tilt implies. We present the replication as evidence on the mechanism, not as a claim that the identical recipe beats every local market.

5.4 Transaction Costs and the When-to-Trade Rule

The results above are gross. Transaction costs are where momentum strategies conventionally die, and net-of-cost evaluation is the standard a trading claim must meet (Novy-Marx and Velikov, 2016; Detzel et al., 2023). Because the strategy's entire active return is concentrated in panic months, its economic case is a transaction-cost one: the regime signal identifies the minority of months in which trading is worth paying for, and the analysis below tests whether relocating turnover to those months converts a cost-fragile strategy into a cost-robust one.

Measured costs

The relevant fee scale for this book is small. Measured from closing quotes across the point-in-time top-1000 universe, the median one-way half-spread is 1.3 basis points in calm months and 1.5 in panic months (90th percentiles 3.5 and 4.0), so the flat 10 bp convention common in the momentum literature overstates the spread bill for a large-cap book by roughly a factor of seven.[21] Costs still matter, because the strategy's gross active return is itself small: at a flat 10 bp the monthly-rebalance implementation keeps an information ratio of only 0.06, with a breakeven one-way cost of 14 bp (Table 10).

Where the turnover is versus where the payoff is

The cost problem has a regime structure. The 134 calm months carry roughly three-quarters of the book's total trading while earning +0.04% per month of active return; the 45 panic months earn +0.72% per month. A monthly-rebalanced implementation therefore pays for turnover all year to collect a payoff that arrives in a quarter of the months, which is exactly the configuration in which trading costs destroy a strategy.

[21] Closing-quote spreads are a lower bound in fast markets, and at institutional size the binding constraint becomes market impact rather than quoted spreads; the flat-cost grid below (up to 50 bp) can be read as spanning those frictions.

Not trading in panic is free; not trading in calm is profitable

Two counterfactuals make the point sharp. First, suppressing rank-driven sells during panic months, holding existing positions through the panic rather than liquidating on rank, *raises* the gross information ratio from 0.21 to 0.33 and nets 0.22 at 10 bp: the no-sell rule is not merely a cost saving but a performance improvement, consistent with the deep-crisis selection mode of Section 5.2.2. Second, the panic sells themselves buy nothing: stocks sold on rank during panic months went on to perform within half a percentage point of their replacements over the following three months and within 0.2 points over six, a wash against any positive trading cost. This is a portfolio-level rule, not a stock-picking one: buying the very deepest losers in panic does not work at the individual-stock level (equal-weighted, the deepest-loser panic entries return roughly -0.3% over three months), and the panic payoff comes from the cap-weighted book tilt with monthly refresh, not from bottom-fishing single names.

Execution policy	Turnover (%/mo)	Gross IR	Net IR @10bp	Breakeven (bp)
Monthly full rebalance	75.7	0.21	0.06	14
Panic no-sell overlay	55.7	0.33	0.22	29
Best static band (40,40)	49.7	–	0.29	35
Regime band (40/15)	54.1	–	0.39	42

Table 10: Execution-policy ladder for the main strategy (one-way turnover; net figures at a flat 10 bp one-way cost; breakeven is the one-way cost at which net active return reaches zero). Banding follows the buy/hold-spread construction: a stock enters the long portfolio at the top decile and is held until its score rank falls below the exit threshold E; the regime band (calm $E = 40$, panic $E = 15$) tightens the exit threshold in panic months only. The (10,10) band reproduces the monthly rebalance exactly.

A regime-conditional trading band

The when-to-trade logic generalizes into an execution rule. Under a buy/hold-spread band (Novy-Marx and Velikov, 2016, 2019), positions enter at the top decile and are held until their rank decays through an exit threshold; widening the band cuts turnover at the cost of holding stale positions, a discrete cousin of the optimal partial-adjustment trading of Gârleanu and Pedersen (2013). A single fixed band already helps: the best static band nets an information ratio of 0.29 at 10 bp. Conditioning the band on the regime, wide in calm ($E = 40$) to let winners run and avoid paying for noise trades, tight in panic ($E = 15$) to preserve the re-ranking that panic months reward, moves the frontier to 0.39 at 10 bp with a breakeven of 42 bp, roughly thirty times the measured half-spread. Against the monthly rebalance the improvement is statistically significant (paired CI on net active return $[+0.011, +0.053]$, $t = 2.99$); against the best static band it is a consistent frontier lead whose confidence interval includes zero ($[-0.001, +0.025]$), so the claim is that the regime band *leads* the frontier, not that it dominates it. The result also bears on a pessimistic recent finding: Azevedo et al. (2024) report that standard cost-mitigation techniques generally fail to rescue machine-learning strategies net of costs. On this book, banding does preserve the net value, and the regime-conditional band leads the mitigation frontier. The direction is robust: the tighter-in-panic corner beats its mirror image in all ten mirror pairs (sign test $p = 0.001$), is the best cell at every cost level from 0 to 50 bp, and is insensitive to the panic threshold (0.38–0.39 net IR for thresholds 0.4–0.6).

Honest limits

The band grid is chosen in-sample on the 179-month window, and the value is time-concentrated: in the 2011–2017 half, which contains few panic months, no execution policy earns positive net active return, and effectively all of the net value accrues over 2018–2025. The strategy's absolute net active return over the index is not statistically significant ($t = 1.40$); what is significant is the implementation claim, that conditioning trading on the regime beats unconditional implementations of the same signal. That is the deliverable: the drawdown signal tells a large-cap momentum book *when* trading is worth paying for, it identifies the months in which not selling is free money, and it converts a cost-fragile monthly strategy (breakeven 14 bp) into a cost-robust banded one (breakeven 42 bp) using information available in real time.

6 Conclusions

This paper asked when a large-cap, long-only momentum portfolio is worth tilting away from its benchmark, and answered it with a two-stage framework: a drawdown-only probabilistic regime signal routes a nonlinear cross-sectional model between classical winner selection in calm markets and term-structure-wide reversal at panic peaks. The honest headline is that the strategy earns no unconditional risk-adjusted alpha in this configuration; its value is conditional and implementational. Essentially all benchmark-beating return arrives in the quarter of months the regime signal classifies as panic, and the signal's practical product is a when-to-trade rule that concentrates turnover where the payoff is.

6.1 Conclusions and Contributions

Cross-sectional momentum's regime-dependent structure is real and admits a concrete characterization. The regime signal acts as a context variable, conditioning how a nonlinear cross-sectional model uses the full momentum term structure rather than predicting returns directly: it carries roughly five times its equal-weight share of the model's attribution, and once it saturates in panic the momentum features take over the stock-by-stock differentiation (Section 5.2.1). The analysis makes three contributions.

First, the methodological combination of a continuous probabilistic regime filter with a nonlinear cross-sectional model remains novel relative to prior regime-conditional momentum work, which paired simpler regime treatments (binary indicators, threshold-based blends, or fast-slow weights) with linear or rank-based cross-sectional rules. The applied configuration sharpens the recipe to its parsimonious core: a univariate hidden Markov model on market drawdown, chosen on literature grounds and confirmed by a train-only selection rule that never adds a second stress feature (Section 4.1).

Second, the empirical analysis characterizes how the cross-section reorganizes across regimes at the level of individual stock-month picks. A four-cluster decomposition of the model's own selections traces the cycle from strong winner selection in calm ($+1.2\,\sigma$ above the cross-sectional mean in the continuation mode) to term-structure-wide reversal at panic peaks ($-1.0\,\sigma$ below, negative at every one of the twelve horizons), a composition no single-horizon rank-based strategy can produce. Cooper et al. (2004) noted this state-dependent asymmetry descriptively from a binary backward-looking market state; this paper implements the regime-conditioned response as a single nonlinear model with a fully-invested long-only book, composition rather than exposure absorbing the regime change.

Third, and most useful to a practitioner, the regime signal converts into an execution rule. The strategy's active return is panic-concentrated while roughly three-quarters of a monthly-rebalanced book's trading occurs in calm months that earn nothing. A fixed trading band alone raises the net information ratio at 10 bp costs from 0.06 to 0.29; conditioning the band on the regime, wide in calm and tight in panic, extends the frontier to 0.39 and the breakeven cost from 14 to 42 bp, a lead over the best static band that holds at every cost level but whose final increment is not individually significant (Section 5.4). To our knowledge no prior work implements a state-dependent buy/hold band on a cross-sectional selection book: the theory literature anticipates state-dependent no-trade regions (Jang et al., 2007; Lynch and Tan, 2011) but has not taken them to portfolio data, and the closest empirical antecedent conditions trading speed, not a no-trade band, on the regime (Collin-Dufresne et al., 2020). The companion finding is that suppressing rank-driven selling in panic is, in point estimate, not merely cheaper but better gross of costs (information ratio 0.21 to 0.33). Together with the calm-month evidence that the signal keeps the book benchmark-like when the tilt earns nothing, the regime signal's deepest contribution is discipline about when not to trade.

6.2 Limitations

Reversal-failure vulnerability and test-window scope

The strategy's primary vulnerability is a prolonged economic deterioration in which the prior decline reflects genuine business failure rather than temporary mispricing, so the beaten-down stocks the model selects do not recover. The panic-mode bet works in short panics but fails in a sustained bear: in the 2000–2002 dot-com bust the extended backtest loses 65.6% against the benchmark's 36.4%, setting the full-sample maximum drawdown of 77.3% (Section 5.3.3). The regime classification was correct in that episode; what changed was the panic-regime pattern, with businesses failing rather than prices being temporarily depressed, and the model cannot distinguish these cases ex ante. The 2011–2025 production window avoids this failure mode, and the 2000–2025 extension documents it exactly once, so we cannot quantify how the strategy would behave across the distribution of such episodes.

A test window containing a further prolonged bear market would deliver a different point estimate. The exposure-scaling overlay discussed in Section 6.3 addresses the structural vulnerability as a circuit breaker. A depression-style scenario lies outside the 25-year backtest, and it is hard to assess via simulation: an accurate stress test would have to model the heterogeneous business-failure dynamics that drive the failure mode, not just shock all stock returns uniformly by a chosen magnitude over a sustained period.

Regime stationarity under an expanding window

The regime model is re-estimated each year, but on an expanding window in which the 1990s and 2000s permanently dominate the training sample. If the market undergoes structural change (a persistent shift in baseline volatility from algorithmic trading, or a different monetary-policy regime), thresholds anchored to decades-old drawdown behavior may miscalibrate, classifying structurally calm periods as panic. The sustained elevation of π_t^{filter} in parts of the recent sample may partly reflect such drift rather than genuine prolonged stress. A rolling estimation window, or online updating that discounts distant history, would address this directly, at the cost of noisier parameters.

In-sample choices and statistical power

The regime indicator itself is protected against specification search: drawdown is fixed on literature grounds and the data-driven check uses training information only (Section 4.1). The residual in-sample choices sit elsewhere. The execution-band grid is selected on the full 179-month window, the candidate pool behind the selection rule was assembled with knowledge of pre-2011 research, and the net value of every execution policy is concentrated in the 2018–2025 half of the sample. Statistical power is the binding constraint throughout: with 45 panic months, the panic-subsample alphas are economically large under some classifications and insignificant under all of them, and the information-ratio differences across ablation variants are directional rather than individually significant. The claims the paper rests on are the ones that clear inference in their exact form: the banded implementation beats monthly rebalancing of the same signal ($t = 2.99$), and the tighter-in-panic direction beats its mirror in all ten pairs (sign test $p = 0.001$). The regime band's lead over the best static band, the panic-subsample alphas, and the claim of beating the index outright ($t = 1.40$) do not clear inference; the panic concentration of active return is economically unambiguous but is documented without a formal subsample test.

6.3 Recommendations for Future Research

Five directions extend this framework most directly.

Richer regime-signal representation

The current model uses only the contemporaneous filtered probability π_t^{filter}, discarding regime dynamics. Two complementary extensions follow. The first looks forward, computing a k-step regime forecast $P(s_{t+k} = 1 \mid \mathbf{z}_{1:t})$ from the HMM's transition probabilities so the portfolio can reposition before a regime shift rather than after it. The gain is concentrated in transition months, where the current framework trades on a stale view of the regime. The second looks backward, feeding the trajectory of π_t^{filter} rather than its current value so that a fresh spike (where mean reversion is most likely) and a sustained elevation (where the reversal mechanism may be breaking down) become distinguishable. A sequential model, or simple lag-based feature engineering (lagged π_t^{filter} values, a short-window rate of change, a regime-duration counter), would expose this trajectory. Together, the two extensions provide anticipation and memory, and the trajectory features bear directly on the reversal-failure limitation above.

Alternative nonlinear models

The selection rule found here is what XGBoost can learn from the regime probability and the twelve momentum horizons; a more expressive model may recover a sharper version of the same signal. The most natural alternative is a feedforward neural network, which Gu et al. (2020) find competitive with tree ensembles for cross-sectional return prediction. Beyond raw performance, the robustness question is the interesting one: if the regime-conditional selection shift recurs in a non-tree architecture, the mechanism is a property of the data rather than an artefact of tree splits, and comparing the network's per-feature attribution to the SHAP decomposition in Table 6 would reveal whether the regime signal plays the same routing role outside the tree representation.

Beyond the long-only mandate

The applied study deliberately excludes the short side, where large-cap borrow is cheap but the crash risk of shorting panic-month rebounds is the classic momentum failure mode. The short side need not be a mirror image of the long leg. Whether a modest short overlay adds value net of borrow costs and crash risk in a large-cap universe, and whether the regime signal should gate the short book more aggressively than the long book, are open implementation questions.

Regime-conditional feature sets

Section 5.1.3 shows that adding firm-level fundamentals inside the regime-conditioned model destroys its panic-month active return while a momentum-plus-fundamentals model without the regime signal remains competitive, suggesting the optimal feature set is itself regime-dependent. A layered construction in which the model produces the ranking and fundamentals enter as a post-ranking quality filter on the long book is a natural starting point; a more ambitious variant trains separate cross-sectional models for the calm and panic branches, each with its own feature set. Both designs require careful out-of-sample validation because the regime-conditional split reduces effective sample size.

Exposure-scaling overlay

An exposure-scaling overlay in the spirit of Daniel and Moskowitz (2016), layered on top of the stock selection, would provide a circuit breaker against sustained downturns that violate the panic-rebound premise. The overlay requires a triggering rule; candidates include π_t^{filter} above a threshold for a minimum number of consecutive months, or its rolling maximum over a fixed window exceeding a calibrated bound. The 2011–2025 production window cannot discriminate among candidate rules, because it contains no prolonged bear market and any overlay only reduces exposure in months that turn out to be profitable; the dot-com sub-period of the 2000–2025 extension (Section 5.3.3) is the appropriate setting to evaluate them, and a rule that converts even part of that episode's −65.6% into benchmark-like performance would remove the strategy's worst-case objection.

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Scope note

Unless labeled otherwise, the appendices below document the long-short, full-universe research configuration from which this framework was developed (four-feature regime signal, 200-seed HMM ensemble, NYSE-breakpoint long-short deciles, 2011–2024 evaluation). Methodological material referenced from the main text (estimation and convergence diagnostics, factor-model and SHAP definitions, the risk-aversion utility construction) carries over to the applied configuration unchanged; numerical results quoted in appendices refer to the companion configuration and are superseded, where they overlap, by the applied numbers in the main text.

Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody's BAA and AAA corporate bond yields for computing the credit spread.

Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, -30% is imputed.

Table 11 summarises all data variables.

Variable	Source	Description	Used in
Stock-level variables			
Monthly returns	CRSP `msf`	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP `msf`	For market equity computation	Section 4
Exchange, share codes	CRSP `msf`	Universe filtering (codes 10, 11)	Section 4
Delisting returns	CRSP `msedelist`	Survivorship bias correction	Section 4
Book equity (`ceqq`)	Compustat `fundq`	Book-to-market, ROE	Appendix L
Total assets (`atq`)	Compustat `fundq`	Asset growth, gross profit, CF ratios	Appendix L
Income (`ibq`)	Compustat `fundq`	ROE, earnings growth, accruals	Appendix L
Revenue (`revtq`)	Compustat `fundq`	Gross profitability	Appendix L
COGS (`cogsq`)	Compustat `fundq`	Gross profitability	Appendix L
Debt (`dlttq`, `dlcq`)	Compustat `fundq`	Leverage ratio	Appendix L
Operating cash flow (`oancfy`)	Compustat `fundq`	CFO, FCF, accruals	Appendix L
Capital expenditure (`capxy`)	Compustat `fundq`	Free cash flow	Appendix L
Market-level variables (HMM inputs)			
Daily market returns	CRSP `dsi`	Drawdown (DD) computation	Section 4.2
Market price index	CRSP `dsi`	Drawdown (DD)	Section 4.2
Cross-sect. stock returns	CRSP `msf`	Dispersion (DISP), REL_N	Section 4.2
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.2
Derived features			
$mom_1, \ldots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
10 fundamentals (ablation only)	Computed	bm, roe, gross profit, cash flow, etc.	Appendix L

Table 11: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The applied sample uses momentum lookbacks from January 1991, an initial training window of 1991–2010 (240 months), and an out-of-sample walk from January 2011 to November 2025 (179 months); see Table 1.

Appendix B: HMM Feature Selection Tables

The two tables in this section support the companion configuration's four-pass feature selection, which is distinct from the train-only one-standard-error rule of Section 4.1. Table 12 reports the top 10 candidate feature combinations on ensemble Sharpe and cross-seed stability. Table 13 reports include-one and leave-one-out ablations of the companion four-feature set.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 12: Top 10 feature combinations from the four-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 20 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

HMM Input Features	XGB Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
Include-one (single-feature HMM)		
DD only (market drawdown)	0.66	−0.31
Leave-one-out (three-feature HMM)		
Drop DD	0.51	−0.46
Drop DISP	0.96	−0.01
Drop REL_N	0.43	−0.54
Drop CS	0.96	−0.01

Table 13: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 2) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone is the strongest single-feature classifier but trails the full set by −0.31. Removing DD or REL_N causes the largest leave-one-out drops (−0.46 and −0.54), while DISP and CS contribute marginally to the Sharpe but improve seed stability (Section 4.1).

Appendix C: Portfolio Formation, Transaction Costs, and Performance Metrics

C.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of the strategy's ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s \in \mathcal{L}_t} ME_t^s}, \qquad w_t^{s,S} = \frac{ME_t^s}{\sum_{s \in \mathcal{S}_t} ME_t^s}, \tag{16}$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t, respectively. The long-short portfolio return is:

$$r_t^{\text{LS}} = \sum_{s \in \mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s \in \mathcal{S}_t} w_t^{s,S} \cdot r_t^s. \tag{17}$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

C.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot (\tau_t^L + \tau_t^S), \tag{18}$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2} \sum_s \left| w_t^{s,L,\text{new}} - w_{t-1}^{s,L,\text{prev}} \right|, \qquad \tau_t^S = \frac{1}{2} \sum_s \left| w_t^{s,S,\text{new}} - w_{t-1}^{s,S,\text{prev}} \right|, \tag{19}$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

Transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models' training objectives. The cross-sectional models (DET, LR, and XGB) optimize stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., Fama and French, 2015; Goulding et al., 2023), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimizes stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimization layer, and remains a potential extension of this work.

C.3 Performance Metrics

Standard Metrics

Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^{n}(1+r_t)\right]^{12/n} - 1. \tag{20}$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12}. \tag{21}$$

Sharpe ratio The annualized Sharpe ratio:[22]

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12}, \tag{22}$$

where $\bar{r}$ is the mean monthly return. Sharpe is computed from the arithmetic monthly mean annualized by $\sqrt{12}$, while annualized returns reported elsewhere in this paper are compound (geometric); under high volatility the two definitions of average return differ, so Sharpe should not be inferred from a table by dividing annualized return by annualized volatility (e.g. at depth 1 in the methodology hyperparameter sweep, the reported Sharpe is 0.53 but 8.7%/19.4% = 0.45).

Maximum drawdown Let $C_t = \prod_{\tau=1}^{t}(1+r_\tau)$ denote the cumulative wealth path. The maximum drawdown is:

$$MDD = \min_{1\le t\le n} \frac{C_t - \max_{1\le\tau\le t} C_\tau}{\max_{1\le\tau\le t} C_\tau}. \tag{23}$$

Regime-Conditional Evaluation

To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \ge 0.5$) periods, and the Sharpe ratio is computed separately within

[22] All Sharpe ratios use raw returns. The risk-free rate averaged ∼2% over 2011–2024 (near zero pre-2021), so absolute Sharpe ratios are overstated by roughly 0.05–0.10; relative comparisons are unaffected because the convention is uniform.

each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \qquad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12}. \tag{24}$$

This decomposition helps assess whether the strategy's alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Per-Cluster Sharpe

The cluster Sharpes reported in Section 5.2.2 apply the standard Sharpe formula to the months in each cluster. Let M_C denote the set of months assigned to cluster C, let r_t be the long-short monthly portfolio return at month t, and let $n_C = |M_C|$. The point-estimate cluster Sharpe is

$$\text{Sharpe}_C = \frac{\bar{r}_{M_C}}{\sigma(r_{M_C})} \cdot \sqrt{12},$$

where

$$\bar{r}_{M_C} = \frac{1}{n_C} \sum_{t \in M_C} r_t, \qquad \sigma(r_{M_C}) = \sqrt{\frac{1}{n_C - 1} \sum_{t \in M_C} (r_t - \bar{r}_{M_C})^2}.$$

The denominator is therefore the unbiased sample standard deviation of the in-cluster long-short monthly returns; annualisation by $\sqrt{12}$ follows the monthly-to-annual convention used throughout this paper. Confidence intervals come from a block bootstrap (block size 6, 5,000 reps) on the in-cluster return series.

Appendix D: Feature Importance Methods

D.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP values (Lundberg and Lee, 2017) are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x}) \Big], \tag{25}$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarized as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)|. \tag{26}$$

Two aggregations of I_j appear in the main text. The first is the *total* $|SHAP|$ *share*, each feature's contribution as a fraction of overall importance:

$$\text{share}_j = \frac{I_j}{\sum_{k=1}^{13} I_k}. \tag{27}$$

The 46% figure reported for π_t^{filter} in Section 5.2.1 is share_π computed across the full test set.

The second is the *per-horizon momentum share by leg* plotted in Figure 5, which restricts attention to the twelve momentum horizons within a single leg-regime cell. For leg $\ell \in \{\text{long}, \text{short}\}$, regime $r \in \{\text{calm}, \text{panic}\}$, and horizon $h \in \{1, \ldots, 12\}$,

$$s_{\ell,r,h} = \frac{\sum_{i\in\mathcal{I}_{\ell,r}} |\phi_h(\mathbf{x}_i)|}{\sum_{h'=1}^{12} \sum_{i\in\mathcal{I}_{\ell,r}} |\phi_{h'}(\mathbf{x}_i)|}, \tag{28}$$

where $\mathcal{I}_{\ell,r}$ is the set of test observations in leg ℓ and regime r. By construction the twelve shares within each (ℓ, r) sum to one. Pooling across regimes weighted by observation count gives the leg-level shares displayed in Figure 5:

$$s_{\ell,h} = \frac{\sum_r n_{\ell,r}\, s_{\ell,r,h}}{\sum_r n_{\ell,r}}, \qquad n_{\ell,r} = |\mathcal{I}_{\ell,r}|. \tag{29}$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal's influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix E: Alternative Training Targets for XGBoost

Section 5.3.2 argues that embedding a risk penalty directly in the XGBoost training target degrades performance, motivating the per-direction utility specification used in the main text. Table 14 shows this degradation is not an artefact of the mean-variance functional form: the same pattern appears under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.107	30.0%	27.1%	−28.3%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.061	22.9%	21.6%	−22.3%
	$\gamma = 1$	1.024	16.5%	16.2%	−22.8%
	$\gamma = 2$	1.135	15.5%	13.6%	−22.0%
	$\gamma = 5$	1.062	13.5%	12.7%	−22.5%
	$\gamma = 10$	1.082	13.1%	12.1%	−18.9%
Log mean-variance: $\log(1{+}r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.051	14.6%	13.9%	−17.9%
	$\gamma = 1$	1.039	13.8%	13.4%	−23.8%
	$\gamma = 2$	1.080	14.0%	12.9%	−20.9%
	$\gamma = 5$	1.127	14.1%	12.4%	−18.7%
	$\gamma = 10$	0.995	12.1%	12.3%	−20.2%
Sharpe-like: $r \,/\, \sigma^a$	$a = 0.5$	1.148	24.8%	21.4%	−22.4%
	$a = 1.0$	0.984	14.6%	15.0%	−29.1%
	$a = 1.5$	0.936	12.4%	13.5%	−21.0%
	$a = 2.0$	0.759	12.8%	18.1%	−24.5%

Table 14: Out-of-sample performance of XGBoost (XGB) under alternative training targets. The construction here is long-only (top-decile, NYSE-breakpoint, value-weighted), used as a diagnostic that isolates the effect of the training target itself from the long-short construction; the absolute Sharpes are therefore not comparable with the long-short headline in Table 2, only the relative ordering across rows. Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the diagnostic in Section 5.3.2 is not specific to the mean-variance utility formulation; the single-model training-target approach is structurally unsuited to incorporating risk aversion in a long-short setting, which the long-and-short utility construction resolves.

Appendix F: Long-Short Utility Construction

This appendix supplements the long-and-short utility specification in Section 5.3.2 with the volatility-forecast model, the high-γ collapse mechanism, and the sweep grid.

Volatility forecast

The expected-return forecast $\hat{r}_t^s$ comes from the XGBoost ensemble described in Section 3. The volatility forecast $\hat{\sigma}_t^s$ comes from a second XGBoost ensemble trained on the realized 12-month forward volatility

$$\sigma^s_{[t+1,\,t+12]} \;=\; \mathrm{std}\big(r^s_{t+1}, \ldots, r^s_{t+12}\big),$$

using the same feature set augmented with trailing 12-month realized volatility. The 50-seed ensemble is trained on 1990–2010 and evaluated out-of-sample on 2011–2024.

Convergence to dollar-neutral noise at high γ

Stocks appearing in both top deciles of u_{long} and u_{short} (the overlap set, which grows with γ) are retained on both sides, and their contributions to the long and short books net out in $r_L - r_S$. This is the mechanism by which the long-short spread converges to zero as γ rises beyond the useful range. At high γ, the volatility penalty dominates the return signal, low-$\hat{\sigma}$ stocks occupy both top deciles simultaneously, and the two legs become near-identical value-weighted low-volatility portfolios.

Sweep grid

The risk-aversion sweep in Figure 7 uses a 16-point grid over $[0, 1]$: step 0.05 up to 0.5, step 0.1 thereafter.

Appendix G: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

G.1 Sub-Period Stability

To assess whether the main findings are driven by a single favorable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2024.

	2011–2015	2016–2020	2021–2024	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
LR	0.47	-0.02	-0.38	-0.01
XGB	0.59	1.04	1.70	1.11

Table 15: Sub-period Sharpe ratios (long-short). The test period is split into three roughly equal sub-periods. XGB is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, XGB is the only strategy that produces positive Sharpe ratios in every sub-period (0.59, 1.04, 1.70). LR collapses in 2016–2020 (Sharpe -0.02) and turns negative in 2021–2024 (-0.38). Fixed 12-month momentum turns negative after 2015, consistent with the momentum crash mechanism documented by Daniel and Moskowitz (2016). XGB's strongest performance (1.70) is in 2021–2024, a period characterized by elevated volatility and multiple stress episodes, precisely the conditions where regime-conditional stock selection adds the most value.

G.2 Turnover and Transaction Cost Sensitivity

A strategy's net-of-cost performance depends critically on its turnover. Table 16 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
LR	168.8%	2,026%
XGB	132.3%	1,588%

Table 16: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly × 12).

In long-short construction, XGB's average monthly one-way turnover is 132.3%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 17 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.06	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
LR	0.12	0.05	-0.02	-0.16	-0.29	-0.57
XGB	1.19	1.15	1.11	1.03	0.95	0.78

Table 17: XGB remains profitable at 50 bps. All other strategies become unprofitable at moderate cost levels.

XGB remains profitable across all cost levels tested. Even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.78. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

G.3 Model Sensitivity

The XGBoost hyperparameters are set on the training period and stress-tested out of sample; tree depth in particular is analysed in Appendix G.4. To verify that the results are not an artefact of a particular configuration, Table 18 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=1, lr=0.05, n=500	0.53	8.7%	19.4%
depth=2, lr=0.05, n=500	0.86	15.7%	19.1%
depth=3, lr=0.05, n=500	0.98	19.2%	19.9%
depth=4, lr=0.05, n=500	1.11	21.7%	19.5%
depth=5, lr=0.05, n=500	0.92	17.4%	19.4%
depth=6, lr=0.05, n=500	0.90	17.0%	19.5%
depth=4, lr=0.01, n=500	0.99	18.7%	19.2%
depth=4, lr=0.10, n=500	1.00	19.6%	19.9%
depth=4, lr=0.05, n=200	1.06	20.5%	19.4%
depth=4, lr=0.05, n=1000	0.96	18.7%	19.9%

Table 18: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

Hyperparameter sensitivity

At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance peaks at depth 4 and falls off on either side (see Appendix G.4 for the depth analysis). The results confirm that XGB's performance is not an artefact of a particular hyperparameter configuration.

G.4 Tree Depth and Interaction Order

The maximum tree depth is the order of feature interactions the model can represent: a decision path of length d conditions on d features jointly. Depth therefore controls whether the regime signal interacts with single momentum horizons or with the joint configuration of several. Depth 4 is the smallest order at which one path can gate on the regime (depth 1) and then combine three momentum horizons (depths 2 to 4), a regime-conditioned read of term-structure shape rather than of one horizon in isolation. This is the interaction the mechanism of Section 5.2.2 turns on.

Figure 8 reports two views of the depth response, both at the production hyperparameters (learning rate 0.05, 500 trees). The left panel is a five-fold expanding-window cross-validation on the training period (1990–2010), each fold trained on an initial window and validated on the following two years, scored by net-of-fee long-short validation Sharpe. The right panel is the out-of-sample sensitivity sweep on the 2011–2024 test window (Table 18). In cross-validation the depth response is flat: validation Sharpe sits between 0.44 and 0.52 across all six depths, a spread far smaller than the across-fold standard deviation of roughly 0.7, so no depth is statistically separable from another in sample. Allowing each depth its own best learning rate and tree count does not change this: the best per-depth configurations still fall within fold noise of one another, and the single best configuration in the entire grid lies at depth 3 (validation Sharpe 0.60 against

0.58 at depth 4), a lead of 0.02 that is negligible against the same noise. Cross-validation cannot resolve the depth.

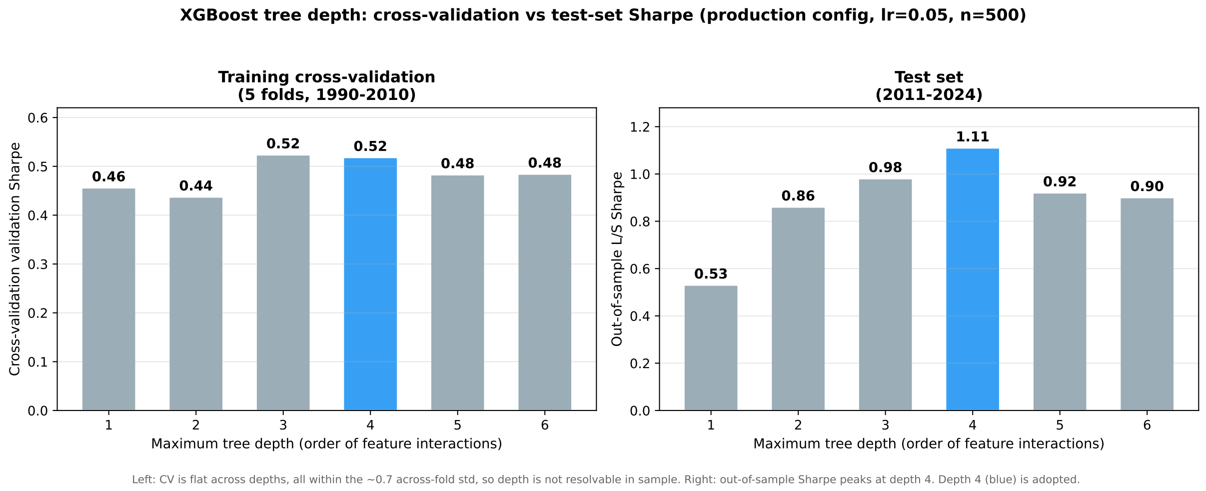

Figure 8: XGBoost tree-depth response at the production hyperparameters (learning rate 0.05, 500 trees). Left: five-fold expanding-window cross-validation validation Sharpe on the training period (1990–2010); the response is flat across depths, all within the across-fold standard deviation of roughly 0.7, so no depth is separable in sample. Right: out-of-sample long-short Sharpe on the 2011–2024 test window, which peaks at depth 4. Depth 4 (blue) is the adopted depth.

On the evaluation window the depth response is more pronounced, though this is the same window on which the headline result is reported: out-of-sample Sharpe rises from 0.53 at depth 1 to a peak of 1.11 at depth 4, then falls to 0.92 and 0.90 at depths 5 and 6. This rise is consistent with the interaction-order account above, the third within-regime level adding signal, while the decline beyond depth 4 is consistent with the added order capturing noise rather than signal. We adopt depth 4 because it is the smallest interaction order that completes the regime-conditioned term-structure read; the out-of-sample peak corroborates this a-priori choice but, coming from the evaluation window, is treated as corroboration rather than as the selection criterion. The mechanism itself is qualitatively the same across depths 3 and 4, so the choice within that pair does not affect the reported results.

Finally, the depth choice is independent of the number of selection modes recovered in Section 5.2.2. The value $K = 4$ is a K-means partition of the long-leg cross-sectional z-curves, computed on the model's monthly picks and chosen on economic grounds; it is not a property of the tree depth, and the same partition could be applied to the picks of a shallower model. The evidence for the regime-momentum mechanism, namely the SHAP attribution, the z-curves, and the leg-level betas, likewise does not rest on the depth. Depth 4 sets the model's capacity to express the mechanism, not the number of modes the downstream clustering reports.

G.5 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity (for example, distinguishing mild corrections from severe crises) would improve the downstream portfolio models. Table 19 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

	LR		XGB	
K	Mean Sharpe	Std	Mean Sharpe	Std
2	-0.012	0.000	1.107	0.000
3	-0.029	0.004	0.831	0.164
4	-0.029	0.003	0.606	0.291
5	-0.028	0.003	0.724	0.087

Table 19: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). The K=2 row uses the 200-seed production regime signal as the reference point (its Std is zero by construction). For $K = 3, 4, 5$ each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). LR is invariant to K. XGB shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K.

In long-short construction, XGB shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K, and the intermediate states that emerge for $K \geq 3$ lack the persistence needed to provide a stable conditioning signal. With only 240 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way, with higher values indicating more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism, for example a "growth" state characterized by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model's ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension, the degree of market stress, which is the dimension most relevant to momentum's cross-sectional structure.

G.6 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum's annual profitability. To verify that XGB's performance is not driven by a January seasonal, Table 20 recomputes all metrics after excluding the 14 January months from the test period. XGB's Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
Full sample (baseline)				
XGB	21.7%	19.5%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
Excluding January				
XGB	20.9%	19.9%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 20: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Appendix H: External Validity

H.1 Baseline Train/Test Split

Table 21 reports the baseline train/test split performance. Robustness across training windows and prediction horizons is provided separately by the 25-year expanding-window backtest (Appendix H.2 below, also reported in Section 5.3.3), which re-estimates the full pipeline annually and therefore subsumes a continuum of alternative splits.

Train / Test split	N_{test}	LR Sharpe	XGB Sharpe
1990–2010 / 2011–2024 (baseline)	167	−0.01	1.11

Table 21: Out-of-sample performance under the baseline train/test split (long-short, mom+π, 50-seed ensemble). LR collapses while XGB achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2024 test), XGB achieves a Sharpe of 1.11 while LR collapses (−0.01), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

H.2 25-Year Expanding-Window Out-of-Sample Methodology

The historical evidence reported in Section 5.3.3 comes from a separate analysis from the one above: a 25-year out-of-sample backtest in which the full pipeline (HMM + XGBoost) is re-estimated every January, with annual retraining initialized in 2000 and each year's predictions made by a model trained only on data available through the prior December. Each retraining uses the production seed configuration described in Section 3 (200 HMM seeds, 50 XGBoost seeds) and the same long-short construction described in Appendix C.1.

The HMM's panic-state label is fixed by an economically motivated sign convention rather than the data-driven percentile comparison used elsewhere. The convention encodes the known direction of each feature in panic regimes: -1 for drawdown, $+1$ for cross-sectional dispersion, -1 for relative market participation, and $+1$ for the credit spread. The data-driven label-switching procedure used in the main pipeline is unstable on short training windows with few observed crisis months (under 20 months in the crisis windows used by the percentile comparison; see Appendix I), so the fixed-sign fallback activates for retrainings before 2002. From 2002 onward the data-driven procedure resumes, by which point the training window contains the dot-com bust as a calibration anchor. The HMM seed loop is parallelised across CPU cores via `multiprocessing.Pool`; the parallel implementation is bit-identical to the serial reference (verified in the test suite by comparing per-seed pi-filter arrays).

Each retraining produces 12 monthly out-of-sample long-short returns, which are stitched into a single monthly return series; this paper reports the 25-year sub-sample from January 2000 to November 2024 (299 months). Every prediction is strictly causal: the model used for month t saw no information after month $t-1$. The 167-month sub-window from January 2011 onward overlaps the main test period reported in Section 5.1 and serves as a methodological cross-check; expanding-window OOS produces a Sharpe of 0.88 over that window, within the production result's bootstrap interval.

H.3 Factor Model Regressions

To test whether XGB's out-of-sample return is explained by known risk factors, monthly long-short returns are regressed on five standard factor models: CAPM, Fama–French three-factor, Carhart four-factor, Fama–French five-factor, and a six-factor specification that augments FF5 with the momentum factor (UMD). Each regression takes the form

$$r_t^{\text{LS}} = \alpha + \boldsymbol{\beta}^\top \mathbf{f}_t + \varepsilon_t, \tag{30}$$

where $\mathbf{f}_t$ is the factor return vector for the specification in question, $\boldsymbol{\beta}$ measures the strategy's exposure to each factor, and α is the average return earned in excess of what those exposures would mechanically generate. The null hypothesis $\alpha = 0$ states that the factor model fully explains the strategy's returns; rejecting it means the strategy earns return that no linear combination of the factors can replicate. Newey–West standard errors (6 lags) are used throughout to correct for residual autocorrelation and heteroskedasticity in monthly returns. Table 22 reports $\hat{\alpha}$ and its t-statistic under each specification; Table 23 repeats the exercise for the fund-augmented variant. Alphas that survive all five factor models indicate return that cannot be attributed to market, size, value, profitability, investment, or momentum exposure.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.4	4.08	-0.14					
FF3	23.2	4.48	-0.14	+0.04	-0.17			
Carhart	24.3	4.75	-0.19	-0.01	-0.22	-0.20		
FF5	22.9	4.62	-0.13	+0.08	-0.23		+0.06	+0.12
FF6	24.1	4.81	-0.18	+0.01	-0.31	-0.22	+0.01	+0.21

Table 22: Factor model regressions for XGB (mom+π). α is annualised. t-statistics use Newey–West standard errors (6 lags).

For completeness, Table 23 reports the same regressions for the fund-augmented variant (XGB: mom+π+fund) discussed in Section 5.1.3. Alphas remain statistically significant under all five specifications ($t > 2.9$) but are materially smaller than the baseline's, consistent with the dilution documented in the main text.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	17.6	3.04	-0.16					
FF3	17.1	3.12	-0.14	-0.06	-0.13			
Carhart	18.2	3.22	-0.19	-0.12	-0.18	-0.20		
FF5	16.4	3.02	-0.11	+0.01	-0.30		+0.09	+0.35
FF6	17.7	3.17	-0.16	-0.06	-0.39	-0.23	+0.04	+0.44

Table 23: Factor model regressions for XGB (mom+π+fund). α is annualised. Newey–West t-statistics (6 lags).

Residual diagnostics

To verify that the 6-lag Newey–West truncation is empirically appropriate for our data, we apply the Ljung–Box portmanteau test to the residuals of each factor regression. Table 24 reports the Ljung–Box statistic at lag 6 and the p-values at lags 6 and 12 for every (strategy, model) pair; the test supports the 6-lag truncation for every entry in the table ($p > 0.05$ throughout). Figure 9 plots the residual autocorrelation function out to lag 24 for the six-factor regression of each strategy. Individual autocorrelations are small in magnitude; a small number of lags fall marginally outside the $\pm 1.96/\sqrt{T}$ bands (lag 6 for XGB at $\hat{\rho}_6 = 0.154$, for example) but none reaches Bonferroni-corrected significance across 24 lags, and the joint Ljung–Box test fails to reject. As a direct robustness check on the lag truncation, the t-statistic on the XGB six-factor α is stable across Newey–West truncation lags $L \in \{3, 6, 12, 24\}$, taking values $\{4.77, 4.81, 4.88, 4.74\}$. Whatever residual dependence exists at long lags is small enough that inference does not depend on the choice of truncation.

			LB(6)		LB(12)	
Strategy	Model	T	Q_6	p	Q_{12}	p
XGB	CAPM	167	8.23	0.221	15.59	0.211
XGB	FF3	167	9.45	0.150	17.61	0.128
XGB	Carhart	167	10.46	0.106	17.20	0.142
XGB	FF5	167	9.65	0.140	17.72	0.125
XGB	FF6	167	10.74	0.097	17.51	0.132
D&M	CAPM	167	2.94	0.817	5.57	0.936
D&M	FF3	167	2.82	0.831	5.31	0.947
D&M	Carhart	167	2.88	0.824	5.40	0.943
D&M	FF5	167	2.32	0.888	4.79	0.965
D&M	FF6	167	2.18	0.903	4.63	0.969
WML	CAPM	167	6.16	0.406	16.24	0.180
WML	FF3	167	7.46	0.281	18.18	0.110
WML	Carhart	167	6.84	0.336	18.13	0.112
WML	FF5	167	7.13	0.309	17.83	0.121
WML	FF6	167	6.87	0.333	17.82	0.121
LR	CAPM	167	2.67	0.849	18.99	0.089
LR	FF3	167	2.36	0.884	18.90	0.091
LR	Carhart	167	2.34	0.886	19.08	0.087
LR	FF5	167	2.71	0.844	17.71	0.125
LR	FF6	167	2.71	0.844	17.73	0.124

Table 24: Ljung–Box residual diagnostics for the factor regressions in Table 22. Q_6 and Q_{12} are the Ljung–Box statistics at lags 6 and 12, each reported with its p-value. Across every (strategy, model) pair we fail to reject the null of no residual autocorrelation at the 5% level. The Newey–West truncation of six lags used throughout the thesis is empirically supported.

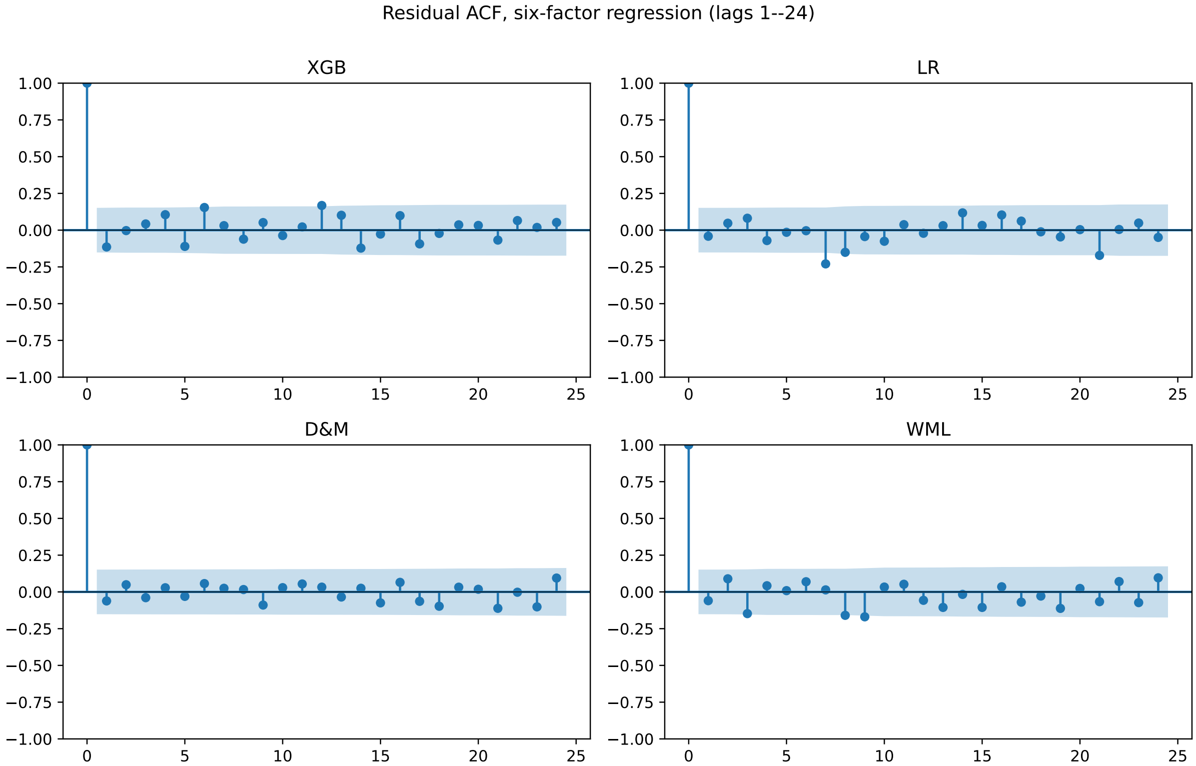

Figure 9: Residual autocorrelation function for the six-factor regression of each strategy. Shaded bands are the $\pm 1.96/\sqrt{T}$ confidence interval under the null of white-noise residuals. Individual autocorrelations are small in magnitude; the occasional lag sitting marginally outside the band does not reach significance after multiple-testing correction, consistent with the joint Ljung–Box result in Table 24.

H.4 Block Bootstrap Inference

To assess the sampling variability of XGB's Sharpe ratio and test its outperformance formally, a block bootstrap is applied to the monthly return series. Returns are resampled with replacement in 12-month blocks to preserve autocorrelation, producing 10,000 synthetic 167-month paths (fixed seed 42). The Sharpe ratio is recomputed on each resample to obtain a 95% confidence interval, and the same block draws are used for XGB and each benchmark to form a paired test of the Sharpe difference. The bottom panel repeats the exercise on XGB returns split by regime.

	Sharpe	95% CI	vs XGB (p)
XGB	1.11	$[0.67, 1.53]$	—
LR	−0.01	$[-0.41, 0.42]$	0.001***
DET (Formula)	0.11	$[-0.36, 0.59]$	0.008***
Fixed 12-mo mom	0.06	$[-0.43, 0.59]$	0.005***
Fixed 1-mo mom	0.26	$[-0.21, 0.73]$	0.021**
XGB regime-conditional Sharpe			
XGB Calm (n =108)	0.84	$[0.29, 1.32]$	—
XGB Panic (n =59)	1.54	$[0.85, 2.22]$	—
Panic − Calm	0.70	[-0.16, 1.59]	0.109

Table 25: Block bootstrap confidence intervals for Sharpe ratios and paired tests of XGB's outperformance. Uses 12-month blocks and 10,000 resamples with a fixed seed. The "vs XGB (p)" column reports two-sided p-values from paired tests where both strategies are resampled on the same dates. $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

XGB's Sharpe confidence interval is [0.67, 1.53], comfortably above zero, while every benchmark's interval straddles zero. The paired tests reject equality of Sharpe ratios between XGB and each benchmark at $p < 0.025$. Within XGB, the panic-period Sharpe exceeds the calm-period Sharpe in point estimate (1.53 vs 0.84), but with only 59 panic months the paired difference is not statistically significant ($p = 0.109$); the 95% interval $[-0.16, 1.59]$ straddles zero. The point-estimate gap is consistent with the regime-conditional selection mechanism described in Section 5.2, but its statistical confirmation would require a longer panic sample. These intervals describe sampling variability within the 2011–2024 sample; out-of-sample generalization is addressed separately by the 25-year expanding-window backtest (Section 5.3.3; methodology in Appendix H.2) and the sub-period analysis (Appendix G.1).

H.5 Leg-Level CAPM Betas by Regime

The D&M comparison in Section 5.3.3 characterizes XGB's contribution as stock selection rather than exposure scaling. A direct test of this framing is to compute leg-level CAPM betas separately by regime: if XGB reorganizes which stocks it holds across regimes (cross-sectional re-ranking), the long-leg and short-leg betas should shift between calm and panic in opposite directions; if instead XGB mechanically scales exposure, both legs' betas should move together.

Table 26: Leg-level CAPM betas by regime, 2011–2024 OOS test period. Panic months are those with $\pi_t^{\text{filter}} \geq 0.5$. Newey–West standard errors (3 lags) in parentheses.

Strategy	Leg	Full	Calm	Panic
Fixed 12-mo mom	Long	1.09 (0.06)	1.09 (0.09)	1.07 (0.08)
Fixed 12-mo mom	Short	1.74 (0.11)	1.80 (0.21)	1.69 (0.13)
XGB	Long	1.55 (0.08)	1.40 (0.11)	1.65 (0.10)
XGB	Short	1.10 (0.06)	1.22 (0.09)	0.98 (0.05)

The pattern in Table 26 is consistent with the cross-sectional re-ranking interpretation. XGB's long-leg beta rises from 1.40 in calm to 1.65 in panic (*tilt amplification*: in panic, the long leg holds names with higher market sensitivity), while the short-leg beta falls from 1.22 to 0.98. Fixed 12-month momentum, by contrast, holds both leg-level betas roughly stable across regimes (long: 1.09 calm, 1.07 panic; short: 1.80 calm, 1.69 panic), consistent with the unconditional rank-based construction. The difference is direct evidence that the regime signal acts on *which* stocks are held rather than on *how much*.

H.6 Per-cluster Feature Signature ($K = 4$ Partition)

Section 5.2.2 reports a $K = 4$ partition of the 167 test months and characterizes each cluster narratively. Table 27 provides the full per-cluster feature signature for completeness: cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster's months; long-leg pick raw momentum at each horizon, averaged across each cluster's months; and long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 7 in the main text.

Feature	Cluster 1 *calm cont.*	Cluster 2 *mild cont.*	Cluster 3 *post-panic*	Cluster 4 *deep crisis*
Cross-section context				
Cross-section mom (overall mean)	13.8%	7.3%	13.5%	−9.3%
Cross-section mom (short tertile, $h = 1$–4)	6.1%	4.4%	3.0%	−7.1%
Cross-section mom (mid tertile, $h = 5$–8)	14.6%	7.6%	12.4%	−11.1%
Cross-section mom (long tertile, $h = 9$–12)	20.8%	9.7%	25.0%	−9.6%
Cross-section dispersion (short)	0.23	0.27	0.28	0.23
Cross-section dispersion (mid)	0.44	0.46	0.56	0.35
Cross-section dispersion (long)	0.60	0.59	0.85	0.45
Cross-section skewness (short)	4.6	7.2	6.4	3.7
Cross-section skewness (mid)	6.2	7.6	6.2	5.6
Cross-section skewness (long)	6.7	8.1	6.5	5.7
π^{filter} frequency, prior 6 months	0.27	0.37	0.30	0.51
π^{filter} frequency, prior 12 months	0.31	0.32	0.41	0.30
Strategy Sharpe, prior 12 months	1.22	1.04	1.15	1.08
Long-leg pick raw momentum by horizon				
Long-leg pick mom, $h = 1$	7.1%	3.1%	−3.4%	−15.3%
Long-leg pick mom, $h = 2$	13.5%	8.6%	−5.5%	−21.6%
Long-leg pick mom, $h = 3$	18.3%	8.8%	−5.3%	−26.3%
Long-leg pick mom, $h = 4$	23.1%	10.6%	−5.8%	−30.8%
Long-leg pick mom, $h = 5$	30.8%	13.8%	−5.9%	−35.0%
Long-leg pick mom, $h = 6$	40.2%	18.0%	−3.2%	−38.3%
Long-leg pick mom, $h = 7$	49.4%	20.7%	−1.3%	−42.1%
Long-leg pick mom, $h = 8$	59.2%	23.7%	0.0%	−44.9%
Long-leg pick mom, $h = 9$	61.3%	23.1%	1.1%	−46.5%
Long-leg pick mom, $h = 10$	60.7%	21.9%	1.3%	−47.2%
Long-leg pick mom, $h = 11$	61.1%	22.8%	1.2%	−47.3%
Long-leg pick mom, $h = 12$	57.9%	18.5%	−5.3%	−47.5%
Long-leg pick cross-sectional z-score by horizon				
Long-leg z-curve, $h = 1$	0.26	0.04	−0.28	−0.73
Long-leg z-curve, $h = 2$	0.38	0.14	−0.31	−0.74
Long-leg z-curve, $h = 3$	0.40	0.11	−0.34	−0.75
Long-leg z-curve, $h = 4$	0.44	0.10	−0.36	−0.80
Long-leg z-curve, $h = 5$	0.54	0.16	−0.35	−0.85
Long-leg z-curve, $h = 6$	0.65	0.23	−0.29	−0.83
Long-leg z-curve, $h = 7$	0.73	0.27	−0.26	−0.85
Long-leg z-curve, $h = 8$	0.80	0.32	−0.22	−0.87
Long-leg z-curve, $h = 9$	0.76	0.29	−0.23	−0.88
Long-leg z-curve, $h = 10$	0.72	0.24	−0.25	−0.87
Long-leg z-curve, $h = 11$	0.67	0.22	−0.26	−0.86
Long-leg z-curve, $h = 12$	0.54	0.12	−0.35	−0.87

Table 27: Per-cluster feature signature for the $K = 4$ partition, 2011–2024 test period. Features are grouped into three blocks: (1) cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each clusterś months; (2) long-leg pick raw momentum at each horizon, averaged across each clusterś months; (3) long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 7; this table is the full descriptive feature signature.

H.7 Cluster-Level Cross-Validation

Three diagnostics extend the regime-aggregate findings of Section 5.2 across the $K = 4$ partition.

Leg-betas across the $K = 4$ partition

XGB retains long-leg $\beta >$ short-leg β in every cluster (spreads +0.68, +0.40, +0.49, +0.43 for clusters 1–4), while unconditional 12-month momentum exhibits leg-beta inversion in every cluster (−0.24, −0.87, −0.75, −0.42; Table 28). Daniel and Moskowitz (2016)'s crash mechanism is latent across every cluster type; XGB escapes it everywhere through compositional re-ranking.

		XGB			Fixed 12-mo mom		
Cluster	n	β_L	β_S	Spread	β_L	β_S	Spread
1 (calm continuation)	53	1.65 (0.22)	0.97 (0.13)	0.68	1.32 (0.22)	1.57 (0.24)	−0.24
2 (transition)	62	1.60 (0.15)	1.20 (0.09)	0.40	1.04 (0.08)	1.91 (0.17)	−0.87
3 (post-panic recovery)	31	1.46 (0.16)	0.97 (0.11)	0.49	1.09 (0.17)	1.84 (0.22)	−0.75
4 (deep crisis)	21	1.49 (0.08)	1.06 (0.09)	0.43	1.07 (0.10)	1.49 (0.14)	−0.42

Table 28: Leg-level CAPM betas by K=4 cluster, 2011–2024 test period. XGB (regime-conditional XGBoost ensemble) versus unconditional 12-month momentum, side-by-side. Standard errors in parentheses. Spread = $\beta_L - \beta_S$.

π_t^{filter}'s SHAP share by cluster

Combined long-and-short SHAP shares are 0.50, 0.46, 0.54, 0.24 for clusters 1–4 (Table 29). These are consistent with the panic-aggregate 41% share reported in Section 5.2.1: the 41% averages across all panic months, and the cluster-level shares show that the collapse is concentrated in cluster 4 (deep panic, share 0.24), exactly where the main-text discussion attributes the decline to π_t^{filter} saturating near 1. The pattern is consistent with the routing interpretation: π_t^{filter} contributes most where the cross-section is ambiguous (cluster 3) and least where it is unambiguously crisis-shaped (cluster 4, where momentum splits alone suffice to produce the loser portfolio).

Cluster	n	Long π share	Short π share	Combined π share
1 (calm continuation)	53	34.0%	50.1%	49.9%
2 (transition)	62	26.9%	46.5%	46.0%
3 (post-panic recovery)	31	27.8%	53.7%	53.6%
4 (deep crisis)	21	11.4%	38.1%	24.4%

Table 29: π^{filter}'s SHAP share by K=4 cluster, 2011–2024 test period. Long and short shares are computed over long-leg and short-leg picks respectively; combined is over all stock-months in the cluster.

Short-leg structure (future work)

The $K = 4$ partition was defined on long-leg z-curves; applying the same labels to the short leg shows the short basket is structurally distinct from a negated long leg, with short-vs-negated-long correlations of +0.94, +0.89, +0.19, and −0.84 for clusters 1–4 (Table 30). The cluster-4 sign flip from positive to negative correlation means the model is not merely shorting the mirror image of its long-leg picks: it selects a qualitatively different basket whose cross-sectional fingerprint moves opposite to a mechanical "short the long-leg pattern" rule. The month-1 short-leg spike in cluster 4 (+0.57) is consistent with Nagel (2012)'s finding that short-horizon reversal returns spike during high-volatility periods: month 1 carries 10.6% of panic short-leg momentum SHAP versus 5.3% in calm. A full short-leg analysis is left to future work.

Cluster	n	z_1	$\bar{z}$	Basket	$\rho(-z^L)$
1 (calm continuation)	53	−0.062	−0.365	391	0.943
2 (transition)	62	0.101	−0.174	334	0.888
3 (post-panic recovery)	31	0.209	0.241	382	0.190
4 (deep crisis)	21	0.568	0.170	353	−0.835

Table 30: Short-leg z-curve summary by K=4 cluster, 2011–2024 test period. z_1 is the cluster-mean cross-sectional z-score of short-leg picks at the 1-month horizon; $\bar{z}$ averages across all 12 horizons. Basket size is mean stocks per month. $\rho(-z^L)$ is the correlation between the short-leg z-curve centroid and the negated long-leg z-curve centroid; values below 0.95 indicate structural asymmetry.

H.8 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student-t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student-t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t{=}k, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−882.7	−701.1	1929.9	91.4%	0.872
5	−829.0	−669.8	1822.5	96.8%	0.960
7	−838.3	−698.2	1841.2	98.0%	0.984
10	−815.8	−644.3	1796.1	96.8%	0.960
15	−800.1	−651.0	1764.8	98.3%	0.997
20	−802.9	−621.7	1770.4	99.3%	0.997
30	−802.6	−640.6	1769.8	98.5%	0.993
50	−796.6	−658.1	1757.7	98.8%	0.996
100	−792.3	−643.5	1749.0	98.8%	0.987
Normal	−787.5	−642.5	**1739.5**	—	—

Table 31: Student-t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student-t at $\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere.

Appendix I: Label Sign Correction

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j, the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\mathrm{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \tag{31}$$

Features that spike upward during crises receive $\mathrm{sign}_j = +1$; features that decline receive $\mathrm{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\boldsymbol{\mu}}_k$ best aligns with the crisis direction:

$$k^* = \arg\max_{k \in \{0,1\}} \sum_{j=1}^{D} \mathrm{sign}_j \cdot \hat{\mu}_{k,j}. \tag{32}$$

The state labeled k^* is designated as "panic" and the other as "calm."

Appendix J: Convergence Diagnostics

Convergence of the Gibbs sampler is assessed in two complementary ways: effective sample sizes (ESS) within each chain, and the Gelman–Rubin $\hat{R}$ statistic across independent chains. The two are complementary because ESS measures efficiency within a single chain but cannot detect whether different chains have converged to different posterior modes, while Gelman–Rubin compares chains but says nothing about within-chain autocorrelation. Trace plots provide a third, visual check.

J.1 Effective Sample Size (ESS)

Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n,

$$ESS = \frac{n}{1 + 2\sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \tag{33}$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ,[23] and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws; an ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

[23] $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\,\hat{\sigma}^2} \sum_{i=1}^{n-\ell}(\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i-th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

J.2 Trace plots

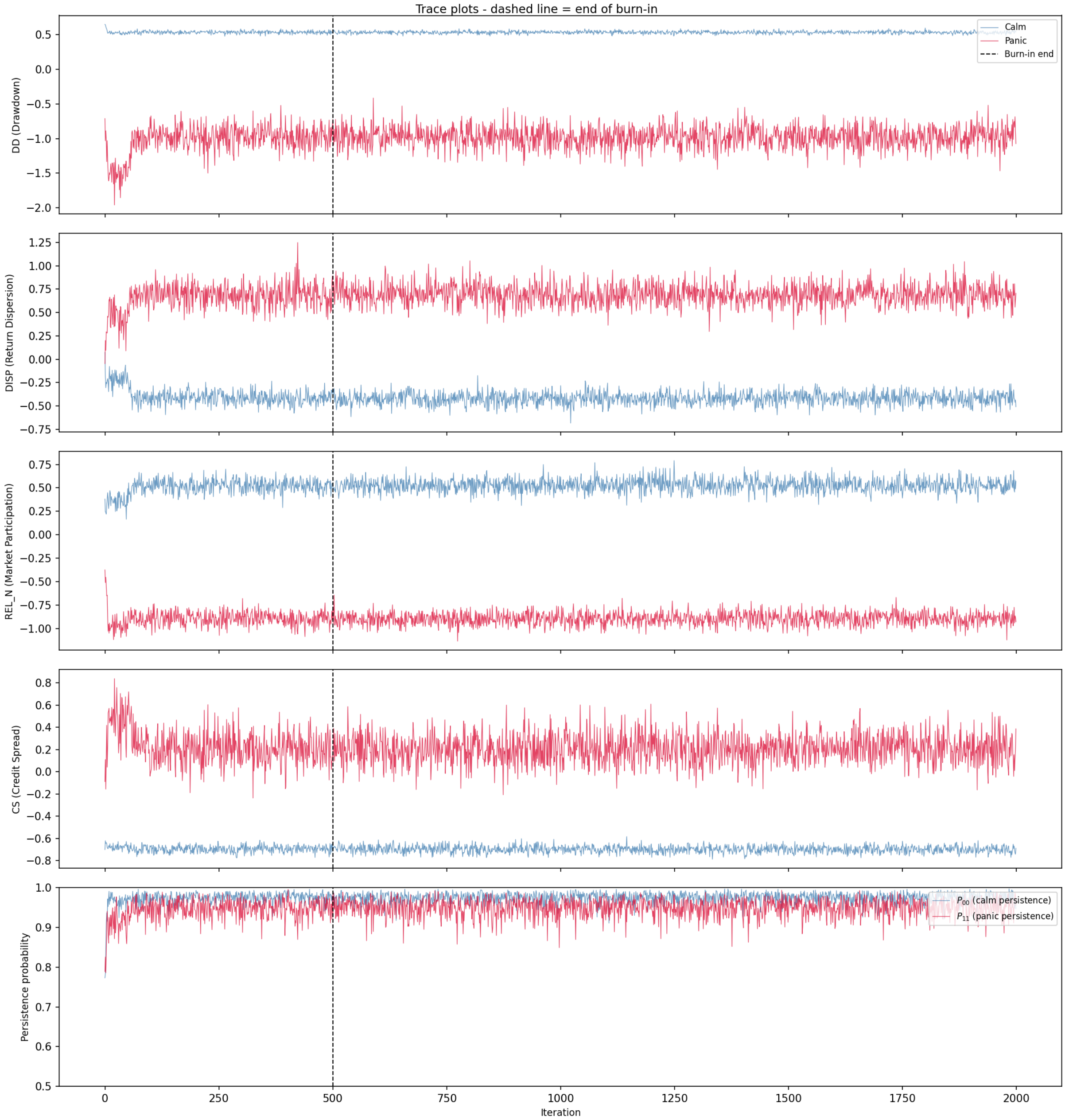

Figure 10: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

J.3 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992; Gelman et al., 2013) compares within-chain and between-chain variance across M independent chains, each of length n.

Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is

$$W = \frac{1}{M}\sum_{m=1}^{M} s_m^2, \tag{34}$$

and the between-chain variance is

$$B = \frac{n}{M-1}\sum_{m=1}^{M}(\bar{\theta}_m - \bar{\theta})^2, \tag{35}$$

where $\bar{\theta} = \frac{1}{M}\sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W. The diagnostic combines these:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n}W + \frac{B}{n}. \tag{36}$$

Values of $\hat{R}$ near 1.0 indicate convergence; the conventional threshold is $\hat{R} < 1.1$, with a stricter modern bound at $\hat{R} < 1.01$.

Parameter	$\hat{R}$	Status
μ_{panic}(DD)	1.000	OK
μ_{panic}(DISP)	1.000	OK
μ_{panic}(REL_N)	1.000	OK
μ_{panic}(CS)	1.000	OK
μ_{calm}(DD)	1.000	OK
μ_{calm}(DISP)	1.001	OK
μ_{calm}(REL_N)	1.000	OK
μ_{calm}(CS)	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\to$ panic)	1.000	OK
P_{10} (panic $\to$ calm)	1.000	OK
P_{11} (panic persistence)	1.000	OK

Table 32: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

J.4 Seed Convergence of the Regime Signal

Gelman–Rubin and the trace plots assess whether a single Gibbs chain has converged to its stationary posterior, but they do not quantify how many chains are needed to stabilize the *downstream* Sharpe ratio. The regime signal π_t^{filter} enters the pipeline as the average of N_s seed-specific forward filters (Section 3.1.5), and because XGBoost is a nonlinear function of π_t^{filter}, MCMC noise in any individual chain does not cancel linearly.

To measure this dependence, we subsample $k \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100\}$

chains without replacement from the seed pool, average the forward filter over those k chains, retrain the XGBoost ensemble on the resulting signal, and record the out-of-sample Sharpe. For each $k < 100$ we repeat on 30 random subsets; $k = 100$ exhausts the pool.

k	n_{sub}	Mean	Median	Std	P25	P75	Min	Max
1	30	1.040	1.031	0.078	0.985	1.090	0.915	1.205
2	30	1.062	1.081	0.069	1.019	1.103	0.894	1.217
3	30	1.049	1.044	0.043	1.014	1.075	0.973	1.143
5	30	1.042	1.042	0.047	1.002	1.083	0.967	1.154
10	30	1.053	1.050	0.036	1.038	1.066	0.965	1.145
15	30	1.075	1.079	0.029	1.055	1.097	1.008	1.139
20	30	1.078	1.080	0.022	1.065	1.090	1.025	1.122
30	30	1.089	1.082	0.025	1.069	1.109	1.052	1.137
50	30	1.093	1.092	0.015	1.082	1.101	1.067	1.124
75	30	1.098	1.102	0.012	1.091	1.105	1.067	1.118
100	1	1.118	1.118	0.000	1.118	1.118	1.118	1.118

Table 33: HMM Gibbs-chain seed convergence of the downstream L/S Sharpe ratio. For each subset size k, we draw n_{sub} random subsets of k HMM seeds from a 100-seed pool, average their forward-filter outputs into π_t^{filter}, retrain the XGBoost ensemble on the averaged signal, and report the out-of-sample L/S Sharpe. The production configuration uses $N_s = 200$ HMM seeds; the 100-seed pool here is the subset over which the experiment is run. Mean Sharpe stabilizes around $k \approx 30$–50; standard deviation across subsets shrinks as $1/\sqrt{k}$.

Two patterns emerge. First, the mean Sharpe rises from 1.04 at $k = 1$ to 1.12 at $k = 100$, so a single-chain point estimate understates the strategy's performance by roughly 0.08 in Sharpe units. The gap closes approximately monotonically (with minor non-monotonic dips at $k = 3$ and $k = 5$ before the trend resumes), and most of the lift is realized by $k = 30$. Second, the interquartile range across random subsets collapses from 0.10 at $k = 1$ to 0.02 at $k = 50$: once $k \geq 50$, any two random subsets of the seed pool produce nearly the same XGBoost portfolio. The production configuration ($N_s = 200$) sits well inside this plateau, which is why the 1.11 Sharpe reported in Section 5 is insensitive to which specific seeds were drawn. The practical implication is methodological rather than cosmetic: a single-chain HMM would not reliably support the main conclusions, so the seed ensemble is a requirement of the design, not a late-stage performance tuning.

Appendix K: International HMM Feature Selection

This appendix documents the *companion long-short study's* international replication, which selected regional feature sets with that configuration's four-pass procedure; the applied replication of Section 5.3.4 instead imposes the univariate drawdown signal per region with no per-region selection (Section 4.5). In the companion procedure, the HMM feature set is selected per region via a quality screen on MCMC convergence and crisis-window alignment, a portfolio-Sharpe ranking on the surviving combinations, a definitive validation at higher seed counts, and an additional stability check across random seed partitions. The international candidate pool comprises the five locally-available stress features (DD, VOL, DISP, REL_N, BANK_REL); BANK_REL is the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress, and substitutes for the US-specific Moody's BAA–AAA spread. With DD required, the procedure evaluates fifteen combinations of size one to four. Both UK and Japan select *DD+VOL+REL_N* as the regional feature set. The transplanted-from-US template (DD+DISP+REL_N+BANK_REL) fails the Pass-1 quality screen for both regions (minimum effective sample size of 14 and 4 respectively), motivating per-region selection. Pass-by-pass CSV results and chosen-features JSON manifests are saved under `results/thesis/` with the prefix `intl_<region>_hmm`.

Appendix L: Fundamental Feature Definitions

Ten firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. Cash-flow items (`oancfy`, `capxy`) are reported year-to-date and are differenced within each fiscal year to recover the per-quarter contribution before the trailing twelve-month sum is taken.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\texttt{revtq} - \texttt{cogsq})/\texttt{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\texttt{atq}_t/\texttt{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\texttt{dlttq} + \texttt{dlcq})/\texttt{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.
- **Operating cash flow over assets (cfo_a):** Trailing twelve-month operating cash flow (`oancfy`, converted to quarterly and summed) scaled by total assets.

- **Free cash flow over assets (fcf_a):** Trailing twelve-month operating cash flow minus trailing twelve-month capital expenditure (`capxy`), scaled by total assets.
- **Accruals:** The accrual component of earnings, $(\texttt{ibq}_{\mathrm{TTM}} - \texttt{oancfy}_{\mathrm{TTM}})/\texttt{atq}$, which captures the non-cash portion of reported earnings.

Fundamentals are winsorized at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds. The cash-flow features (cfo_a, fcf_a, accruals) are available for roughly 83% of stock-months; the remaining seven characteristics exceed 82% coverage individually, and XGBoost handles any residual missingness natively via its default-branch logic.